SALES CONVERSION OPTIMIZATION THOUGH COMBO OFFER ANALYTICS : A CASE STUDY OF SUPERMARKET TRANSACTIONS

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DECLARATION BY THE LEARNER

This is to declare that I have carried out this project work myself in part fulfillment of the PCGBA Program of SCDL.

The work is original, has not been copied from anywhere else and has not been submitted to any other University/Institute for an award of any degree/diploma.

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CHAPTER I

1. INTRODUCTION

1.1 BACKGROUND OF RETAIL SALES ANALYTICS

Retail supermarkets have become one of the fastest growing business formats within the Indian consumer market. They operate on the principle of connecting everyday goods, organised display systems, point-of-sale billing, and structured inventory management into a consistent sales cycle. As urban households increasingly depend on supermarkets for groceries and daily essentials, the role of systematic sales monitoring has gained high importance. Retail sales analytics has emerged as a key discipline through which supermarket data can be translated into meaningful insights on customer choices, price responses, product movement, and commercial performance. The essence of retail analytics lies in examining how goods placed on the shelf are actually converted into billable sales through consumer decisions, pricing patterns, and promotional activities.

In a competitive supermarket environment, each product occupies limited shelf space and competes for consumer attention based on price tags, visual appeal, brand value, packaging, and discount schemes. Conversion of these elements into sales depends on both consumer behaviour and the store's capacity to communicate value. Sales conversion refers to the ability of a supermarket to transform stocked goods into successful transaction records using pricing transparency, visual merchandising, attractive displays, and promotional offers. Combo offers, in particular, represent a widely used promotional mechanism in contemporary supermarkets. They combine two or more complementary products under one discounted value. The purpose of combo schemes is to stimulate higher volume purchases, encourage new product trials, clear slow moving stock, and increase total bill value without pressuring individual product margins.

Combo offers operate on the principle of perceived savings. Consumers are encouraged to purchase larger volumes or try new products because the combined price is lower than the sum of individual prices. When implemented effectively, this strategy benefits both the store and the consumer. The customer receives a better value package, while the

supermarket gains by increasing overall transaction value and accelerating the movement of inventory. Indian supermarkets often use combo offers on processed foods, ready mixes, personal care products, cooking essentials, and cleaning products. This reflects a habit among local households to purchase practical, frequently used goods in cost-effective bundles. Combo strategies also help in balancing stock movement, especially when certain items remain unsold for longer periods.

The rapid shift toward electronic billing systems has further strengthened the importance of sales analytics. Earlier, billing processes were manual and recorded mainly for accounting use. Modern point of sale systems record every product level transaction along with product code, category, price, quantity, discount, and time of purchase. This digital capture of data has changed the way supermarkets observe their own performance. Retail analytics now allows supermarkets to go beyond simple revenue calculations. They can identify which products contribute more volume, which categories require better shelf planning, and which pricing strategies succeed in attracting buyers. With analytics, even small changes in unit prices or quantity preferences can be studied to predict demand trends.

Business Analytics transforms supermarket transactions into evidence driven insights. By analysing actual bill and product data, it becomes possible to measure the effectiveness of combo offers, compare product performance, and link consumer decisions with stock turnover. For instance, if a combo scheme for a cooking ingredient and a kitchen essential generates repetitive purchases, it signals a strong complementary pattern that can be promoted further. If certain items appear frequently in bills despite no discounts, it indicates brand loyalty and price acceptance by customers. At the same time, products that move slowly even at discounted prices suggest weak acceptance among buyers, requiring either rebranding or replacement. These conclusions assist in inventory planning, supply chain management, product ordering cycles, and in determining which goods occupy prime shelf space.

Sales analytics also allows supermarkets to identify the value range of customer purchases. By studying bill size, number of products per bill, and frequency of repeat purchases, supermarkets can classify buyers into budget shoppers, bulk purchasers, and mixed category consumers. Such classification helps in planning seasonal promotions, festival sales, and everyday pricing strategies. For example, budget shoppers may respond better to combo offers on essential items, while premium buyers may repeatedly purchase niche branded goods

irrespective of price changes. These variations highlight how analytics contributes to understanding consumer psychology in a structured manner.

Another aspect of data driven supermarket operations is category level performance analysis. Products in supermarkets are not equal in demand or profit contribution. Essential staples such as rice, pulses, sugar, flour, tea, cooking oil, and everyday toiletries contribute consistently to sales volume. However, some products such as packaged snacks, cosmetics, premium dry fruits, and branded ready mixes contribute more margin per unit. Sales analytics enables balance between volume and margin strategies. A store cannot survive by focusing on margins alone, nor can it rely only on high volume essentials. Analytics enables a realistic mix by visualising how product categories perform across different months and seasons.

The use of time bound data plays an important role in retail analytics. Sales fluctuate according to climate, festival periods, price inflation, crop supply, and shifting consumer priorities. The period between April and September often shows different buying trends compared to the festival months between October and January. By examining data from a period such as 01 April 2023 to 30 September 2025, a supermarket can detect seasonal surges in items such as sweets, spices, cleaning goods, festive snacks, personal grooming products, and dry fruits. It can also observe demand decline in certain categories during monsoon months or during inflation periods. These observations support future ordering strategies and reduce both wastage and stockouts.

Retail sales analytics enables supermarkets to handle vast product ranges, track consumer decisions, and derive practical strategies from factual billing records. The shift to electronic billing has made it possible to read sales as a measurable pattern rather than mere financial transactions. Combo offers, product mix, price strategies, category level analysis, and customer segmentation together create a data driven business model where supermarkets do not depend on guesswork for inventory or promotion. The future of supermarket operations in India is likely to rely heavily on analytical tools that interpret bill wise transactions and convert them into strategic conclusions. This makes retail analytics a vital component of modern business management, connecting operations, inventory, and customer behaviour into a unified system of informed decision making.

1.2 OVERVIEW OF SUPERMARKET DATASET

The retail sector has undergone a considerable transformation in the past two decades, driven by rapid urbanisation, growth in branded food products, and the expanding preference

for organised supermarkets among Indian households. The movement towards structured billing systems in grocery and essential goods markets reflects a shift from loosely documented trade to accountable and traceable consumer transactions. The use of data driven decision making has become increasingly essential for inventory management, price determination, demand prediction, and consumer segmentation. In this context, supermarket datasets have acquired noteworthy value for research in business analytics, retail operations, and consumer purchasing behaviour. The dataset analysed in this study mirrors the operational dynamics of a mid-scale Indian supermarket that handles diverse grocery products across a period of thirty months. It contains primary details on invoicing, product categorisation, item level billing, and sales distribution trends between 01 April 2023 and 30 September 2025. The compilation includes three key files, namely ProductSheet with 544 items, BillMain representing daily aggregated bills, and BillProducts that expands every bill into detailed transactions. Together, these records highlight how a supermarket functions at a micro level with respect to pricing, item diversification, and billing volume.

The ProductSheet.csv file lists a total of 544 items sold during the selected period. The dataset does not limit itself to a narrow product range but reveals a multi-category assortment dominated by household essentials, spices, pulses, cooking ingredients, snacks, cleaning products, and basic personal care goods. The structured representation of each product includes a unique Pcode, the product name, its category, the maximum retail price, and the actual selling price. From a retail analytics perspective, this information provides an important basis for modelling price elasticity, detecting discount trends, and examining the relationship between retail categories and average purchase patterns. The diversity of categories in the ProductSheet shows that Indian consumers frequently combine food staples with other household needs within a single supermarket visit. Another observable attribute is the presence of both domestic and premium priced items. This presents an opportunity to compare consumer purchase responses to value brands and higher priced alternatives. Product names are both in English and regional language scripts, demonstrating that supermarkets adapt labels to local languages to attract a wider consumer base. Thus, the product dataset reflects real market diversity and the inclusive nature of regional retail practices.

The BillMain.csv dataset represents the core of retail operations. It offers bill wise information covering the entire period from 01 April 2023 to 30 September 2025. Within this period, records for 27292 individual bills reveal patterns of customer flow, day-level sales fluctuations, and seasonal variation in purchase quantity. Each record contains a unique BillNo,

date of purchase, the number of products purchased under that bill, the total amount paid, and an indicator showing whether the customer opted for a combo purchase. Combo options, which appear as “Yes” or “No,” are a common feature in contemporary supermarkets. They represent product bundling strategies designed to improve sales volume, encourage impulse buying, and reduce inventory accumulation of specific items. The presence of the combo option in the dataset indicates that customers are not only price conscious but are motivated to choose mixed bundles if they perceive savings or product convenience. The total amount spent in each bill and the number of products purchased are also meaningful variables for analysing cart value trends and product quantity distribution. For example, bills with fewer items but higher total value may indicate premium product purchases, while bills with a large number of low priced items may reflect budget-driven consumption. This dataset therefore creates a foundation for examining supermarket revenue models, consumer psychology, and the impact of promotional strategies.

The BillProducts dataset forms the most detailed layer of the supermarket data. A total of 449978 records reflect every item scanned and billed during the thirty month period. Each row corresponds to a single product transaction, linked with a transaction number, bill number, bill date, product code, product name, product category, the actual rate at which it was sold, the quantity purchased, and the total amount contributed by that item to the bill. Since quantity varies between one and three units, the dataset captures the micro level variation in consumer buying preferences. Basic grocery shopping in India often involves repeat purchases of the same product but in limited quantities. The cluster of repeated lower quantity purchases may indicate that typical customers prioritise affordability per visit, purchase items on demand rather than bulk stock-up, and choose products more frequently than in bigger departmental stores. Another notable aspect in the BillProducts dataset is that transacted product lines can be linked back to their original characteristics in the ProductSheet. This connection enables an extensive range of data driven analyses, including sales forecasting, market basket analysis, item association rules, category level revenue contribution, and price sensitivity modelling.

The combined dataset represents the transactional reality of a medium sized supermarket and reveals characteristic consumption behaviour of Indian households. It also serves as a base for understanding how retail formats operate with digitally recorded billing. Since the period of the dataset overlaps post-pandemic consumption trends, buying behaviour may also reflect cautious spending, re-emerging interest in supermarket shopping, and sensitivity to per unit costs. The extensive presence of fast moving staples such as pulses, rice

varieties, edible oils, basic toiletries, and packaged snacks indicates that frequent purchase products form the backbone of supermarket revenue. This observation aligns with consumer studies which highlight the dominant role of essential goods in sustaining supermarket profitability. The dataset also supports the argument that consumers prefer supermarkets for assured quality, reliable price tags, and variety of brands rather than luxury goods alone.

Another significant aspect is the consistency of product code tracking which makes it possible to evaluate stock movement and identify the most demanded SKUs. When linked with selling price trends, a researcher can also estimate whether customers prefer discounted items or regularly priced items. The ability to compute sales frequency for each of the 544 products over 449978 transactions enables advanced analytical models. These may include predictive analytics for seasonal sales, data mining techniques for uncovering cross-category associations, and optimisation of inventory to avoid overstocking and stockouts. For example, a supermarket can use this dataset to identify which products often appear together in bills, whether a reduction in price of a product leads to higher demand for complementary products, and how combos influence customer choices. Such insights can support store space allocation, targeted discounting, and product bundling.

The presence of a time period from 01 April 2023 to 30 September 2025 demonstrates the importance of temporal variation in retail data. The dataset spans summer, monsoon, and festival seasons over two full calendar years and six months of the third year. Therefore, it is meaningful to track seasonal spikes in items such as sweets, spices, snacks, cleaning goods, or dry fruits during festive months like September to December. A comparison of billing frequency across weekdays and weekends can also provide understanding regarding peak shopping days. If future research integrates external price inflation or commodity supply fluctuations, the dataset could also reveal how market conditions influence consumer choices inside supermarkets.

The dataset structured into ProductSheet, BillMain, and BillProducts provides rich empirical evidence of supermarket functioning. It enables fine grained analysis of customer preferences, product performance, billing patterns, and retail strategy. The breadth of categories, the choice of essential products, and the high volume of individual item transactions illustrate the operational depth of an organised supermarket in an Indian setting. For researchers, this dataset acts as a strong foundation to develop data science models for demand forecasting, market basket learning, dynamic pricing, and sales optimisation. For supermarkets,

it highlights how transaction level data can guide decision making not by assumption but through concrete records of customer purchases. Thus, the dataset portrays a practical, quantifiable, and insightful representation of everyday grocery trade within an increasingly data oriented retail landscape in India.

1.3 RESEARCH PROBLEM

Combo offers have become a familiar promotional practice in supermarkets and are often introduced with the expectation that bundling products will attract customers, increase bill value, and speed up stock movement. Although these schemes appear popular, most supermarkets do not examine how these offers actually influence sales conversion, product preference, revenue generation, or inventory utilisation. Promotional bundles are frequently selected based on experience and store-level assumptions rather than on real transaction patterns. As a result, there is limited clarity on which combos succeed, which items should be paired, and whether bundled discounts contribute to profit or merely generate additional sales volume without value.

Many products that are included in combo schemes do not necessarily convert into individual purchases, making it unclear whether customers buy them due to inherent value or simply because of discounted bundling. Likewise, some slow-moving products continue to remain in stock despite frequent inclusion in combos, suggesting that product selection for offers may not be aligned with actual consumer demand. Supermarkets also do not track whether combo promotions increase the overall basket size or simply shift normal purchases into discounted bills. Without such analysis, there remains a risk of increasing discount dependency instead of improving genuine conversion of products.

Furthermore, supermarkets rarely measure how different product categories perform under combo promotions or whether seasonal demand influences the effectiveness of bundled offers. The absence of category-wise tracking, seasonal evaluation, and product-pair performance limits informed decision making in promotion planning. Equally, there is little understanding of whether discounts support long-term product adoption. It remains uncertain whether customers later purchase the same items individually, or whether they continue to rely only on bundle-based pricing.

Therefore, there is a need to analyse supermarket billing data to evaluate the impact of combo offers on sales conversion, category-level performance, revenue contribution,

customer purchase patterns, and inventory outcomes. Applying business analytics to transaction records can reveal which bundles convert effectively, which products drive revenue, how combos influence basket size, when they perform best, and whether they support long-term purchase behaviour. Such insights are essential for designing evidence-based combo strategies that optimise sales conversion rather than relying on instinctive promotional spending.

1.4 OBJECTIVES OF THE STUDY

1. To measure how many supermarket bills include combo offers and check whether combos are actually converting into customer purchases.
2. To find which products and product pairs appear most in combo bills and identify the items and categories that make combo offers successful.
3. To compare combo and non-combo bills in terms of bill value, quantity bought, profit impact, basket size, revenue contribution, and dependency on discount-based purchases.
4. To recommend better combo offer strategies by analysing slow-moving items, category-wise combo performance, seasonal demand, price sensitivity, and repeat purchases of combo items.

1.5 SCOPE OF THE STUDY

This study uses actual supermarket billing records to evaluate how combo offers influence sales conversion. The scope focuses on:

- Measuring combo conversion rate by identifying how many bills contain combo purchases.
- Finding product and category contribution to combo success by analysing frequently purchased combo items and paired products.
- Comparing combo and non-combo bills in terms of bill value, quantity movement, and revenue impact to check if combos increase profit or only boost sales volume.
- Identifying inventory risk and discount dependency by examining slow-moving products included in combo offers and products that are purchased only through discounts.

- Studying category-wise and seasonal combo performance to understand when and where combo offers should be prioritised.

This study is limited to physical supermarket transactions and does not include online sales, multi-branch comparisons, or consumer demographic profiling beyond billing data.

1.6 DATA REQUIREMENTS

To compute combo-related conversion metrics, the following data must be available from the supermarket billing system:

- Bill-level and item-level records showing product names, quantities, unit price, and total price.
- A clear combo indication in the bill, showing whether items are part of a combo.
- Product price information, including MRP and selling price, to compare profitability between combo and non-combo sales.
- Optional cost or stock data if inventory risk or profit margin analysis is needed for slow-moving or discount-dependent products.

These fields are essential for measuring combo conversion, evaluating product pairing, and analysing performance of combo offers.

1.7 DATA LIMITATIONS

1. Profit and price comparison may be limited if cost price data is not provided, affecting accuracy of combo vs non-combo profitability assessment.
2. Inventory turnover cannot be fully evaluated without detailed purchase, stock-in, and stock-out records, which are needed to measure slow-moving items and stock risk.
3. Consumer buying behaviour cannot be segmented by customer type because demographic details are not part of the billing data.
4. Seasonal patterns may show trends but cannot isolate external influences, such as festival demand, competitor discounts, or price inflation that may affect combo effectiveness.

1.8 METHODOLOGY OF CONDUCT OF RESEARCH

This project uses real supermarket billing records to evaluate how combo offers influence sales conversion, product performance, profitability, and inventory movement. The analysis focuses on actual bill-level and product-level transactions recorded by the

supermarket's billing software. The entire dataset will be processed, analysed, and visualised using simple R-Studio programming.

A. Data Collection Approach

The required data is directly obtained from the supermarket in CSV format. Since combo efficiency must be evaluated using real transactions, the following information is essential:

- **Bill-level details** such as Bill Number, Bill Date, Total Amount, and Combo Status (Yes or No) to measure Combo Conversion Rate and seasonal patterns.
- **Product-level details** such as Product Name, Category, Quantity, MRP, Selling Price, and Combo Indicator to calculate revenue impact, price sensitivity, frequently paired products, and discount dependency.
- **Optional stock or purchase information** if inventory risk or slow-moving product analysis is required.

All files are imported into R-Studio using basic data-loading functions like read.csv().

B. Data Preparation in R-Studio

Once the data is uploaded, it is cleaned, formatted, and made ready for combo analytics using R libraries such as dplyr, tidyr, and readr. Data preparation includes:

- Converting dates and numeric fields into proper formats.
- Removing incomplete or inconsistent rows.
- Creating identifiers for combo items, paired products, and discount-based purchases.
- Structuring data so that product pairs and category-wise insights can be extracted.

C. Derived Calculations Created for Combo Analytics

Certain values are computed during the study because they are not provided directly by the supermarket system. These include:

- **Combo Conversion Rate**

  (Bills with Combo ÷ Total Bills) × 100

- **Combo Discount Dependency Index**

Percentage of products purchased only through combos.

- **Pairing Frequency Counts** for identifying top combo product pairs and strong category combinations.
- **Profit and Quantity Comparison** for combo and non-combo bills, if cost data is available.

These derived calculations help evaluate how combos influence sales, profit, basket size, and repeat purchase behaviour.

D. KPI Computation Using R-Studio

Combo-related KPIs are computed using grouped calculations through functions such as group_by() and summarise(). The key KPIs include:

- Combo Conversion Rate
- Product-Level Combo Contribution
- Frequently Paired Product Combinations
- Profitability Comparison between Combo and Non-Combo Bills
- Inventory Risk Indicator for Slow-Moving Combo Items
- Combo Discount Dependency Index
- Category-Wise Combo Performance
- Basket Size Improvement through Combos
- Revenue Impact of Combo Purchases
- Seasonal Combo Effectiveness
- Price Sensitivity of Combo Products
- Repeat Purchase Influence of Combo Items

These KPIs collectively show whether combo offers increase sales value, improve basket size, support inventory movement, and help in price-driven conversions.

E. Exploratory Data Analysis (EDA) and Visualization

EDA is performed to visualise trends and interpret KPI outputs using R libraries such as:

- **ggplot2** for graphs and charts

- **dplyr** for grouped summaries
- **arules** for identifying frequently paired products

The visual reports include:

- Combo vs Non-Combo bill comparisons
- Category-wise performance charts
- Price sensitivity and dependency charts
- Seasonal pattern visualisation
- Pairing and co-purchase analysis for combo products

F. Software Environment and Scope

All computations are executed only in R-Studio using simple statistical and visual tools. No external machine learning platforms, online tools, application development, or automation systems are used. The study focuses entirely on data interpretation and combo strategy optimisation, based on real-time supermarket billing transactions.

CHAPTER II

2. ANALYSIS OF WORK DONE AND DESIGN

2.1 DATASET PREPARATION

2.1.1 IMPORTING AND UNDERSTANDING THE DATASET

The dataset for this study is drawn directly from the supermarket's billing software. Two primary tables are used. The bill-level table (bill_main) records Bill Number, Bill Date, Total Amount, Number of Products in each bill and whether a combo was opted (Combo_Opted). The product-level table (bill_product) contains Bill Number, Product Code, Product Name, Product Category, Quantity (Number_of_Qty), Selling Price (ProductRate) and the combo flag at item level. Both tables were provided as Excel or CSV exports and imported into RStudio using the readr and readxl packages, after which they were converted into tibbles for easier manipulation.

Understanding these structures is the base for all subsequent analysis. The bill-level table supports indicators such as Combo Conversion Rate, Basket Size comparison and monthly combo trends. The product-level table supports Product-Level Contribution of Combo Purchases, Category distribution, Price Band sensitivity, Frequently Paired Product Combinations, Inventory Risk Indicators, Combo Discount Dependency Index and Repeat Purchase patterns. An initial inspection using glimpse(), summary() and simple frequency tables helped confirm that the fields required for these KPIs are properly recorded and consistent across both datasets.

2.1.2 DATA CLEANING AND STANDARDISATION

Cleaning was carried out to ensure that each transaction contributes correctly to combo analytics. At bill level, duplicate Bill Numbers and obviously repeated records were checked using distinct() and removed where necessary. Bills with missing or invalid dates, or with undefined Combo_Opted values, were excluded so that counts of "Yes" and "No" bills reflect genuine customer decisions.

At product level, duplicates of the same product within the same bill were removed when they appeared as accidental repetitions. Negative or zero quantities were treated as errors

and filtered out. Product names and categories were standardised by trimming spaces, converting to a consistent case and correcting encoding issues such as “Feminine CareÂ” to “Feminine Care”. This standardisation avoids splitting the same product or category into multiple variants and ensures that frequency counts, product-pair extraction and category-wise combo performance are accurate.

A	B	C	D	E
BillNo	BillDate	Number_of_Products	Total_Amount	Combo_Opted
1	01/04/2023	20	2780	No
2	01/04/2023	21	2331	Yes
3	01/04/2023	10	1220	No
4	01/04/2023	10	2290	No
5	01/04/2023	22	5236	Yes
6	01/04/2023	18	4068	Yes
7	01/04/2023	10	1880	No
8	01/04/2023	12	1572	Yes
9	01/04/2023	14	1946	Yes
10	01/04/2023	23	4692	No
11	01/04/2023	28	6524	Yes
27265	30/09/2025	15	3420	No
27266	30/09/2025	23	3013	Yes
27267	30/09/2025	28	5852	Yes
27268	30/09/2025	20	4360	No
27269	30/09/2025	17	3213	Yes
27270	30/09/2025	25	4100	No
27271	30/09/2025	27	6291	Yes
27272	30/09/2025	9	2052	Yes
27273	30/09/2025	23	3703	No
27274	30/09/2025	24	5832	No
27275	30/09/2025	24	5400	No
27276	30/09/2025	12	1644	No
27277	30/09/2025	26	2912	No
27278	30/09/2025	9	1944	No
27279	30/09/2025	20	2260	Yes
27280	30/09/2025	16	3120	No

BillMain

Figure 1. BillMain Dataset

A	B	C	D	E	F	G	H	I	J	K
TransNo	BillNo	BillDate	PCode	Product_N	Product_Category	ProductRa	Number_c	Total_Amc	Combo_Opted	
1	1	01/04/2023	24	Stone Salt	Kitchen Essentials	12	3	36	No	
2	1	01/04/2023	344	Veg Mangc	Fruits & Vegetables	16	3	48	No	
3	1	01/04/2023	541	Whisper U	Feminine CareÂ	569.05	3	1707.15	No	
4	1	01/04/2023	551	Veet Cold	Feminine CareÂ	117.6	1	117.6	No	
5	1	01/04/2023	417	Gopuram I	Household Essentials	21.85	1	21.85	No	
6	1	01/04/2023	139	Maaretti B	Instant Foods	94.05	2	188.1	No	
7	1	01/04/2023	331	Longan Fru	Fruits & Vegetables	156.5	1	156.5	No	
8	1	01/04/2023	276	Banana Ye	Fruits & Vegetables	36.5	2	73	No	
9	1	01/04/2023	506	Savlon Ant	Feminine CareÂ	26.5	3	79.5	No	
10	1	01/04/2023	214	Colgate Ci	Oral Care	14.25	2	28.5	No	
449970	27292	30/09/2025	122	MTR Hing	Masala & Spices	45	1	45	Yes	
449971	27292	30/09/2025	49	Nalam San	Staples	65	1	65	Yes	
449972	27292	30/09/2025	131	Asal Half C	Instant Foods	44.65	1	44.65	Yes	
449973	27292	30/09/2025	537	Fem Oxy B	Feminine CareÂ	53.35	3	160.05	Yes	
449974	27292	30/09/2025	56	Velâ€™s R	Staples	29	2	58	No	
449975	27292	30/09/2025	425	Cycle Om S	Household Essentials	40	3	120	Yes	
449976	27292	30/09/2025	495	Whisper M	Feminine CareÂ	252.2	3	756.6	No	
449977	27292	30/09/2025	434	Cycle Flute	Household Essentials	52	3	156	Yes	
449978	27292	30/09/2025	193	Popular M	Appalam & Papads	97	3	291	Yes	

Figure 2. BillProduct Dataset

Data Preview:

BillNo (double)	BillDate (double)	Number_of_Products (double)	Total_Amount (double)	Combo_Opted (character)
1	2023-04-01	20	2780	No
2	2023-04-01	21	2331	Yes
3	2023-04-01	10	1220	No
4	2023-04-01	10	2290	No
5	2023-04-01	22	5236	Yes
6	2023-04-01	18	4068	Yes
7	2023-04-01	10	1880	No
8	2023-04-01	12	1572	Yes
9	2023-04-01	14	1946	Yes
10	2023-04-01	23	4692	No
11	2023-04-01	28	6524	Yes
12	2023-04-01	17	2023	No

Previewing first 50 entries.

Import Options:

Name: BillMain
Sheet: Default
Range: A1:D10
Max Rows:
Skip: 0
NA:
☑ First Row as Names
☑ Open Data Viewer

Figure 3. Data Preview in R-Studio BillMain

Data Preview:

TransNo (double)	BillNo (double)	BillDate (double)	PCode (double)	Product_Name (character)	Product_Category (character)	ProductRate (double)	Number_of_Qty (double)	Total_Amount (double)	Combo_Opted (character
1	1	2023-04-01	24	Stone Salt 1Kg	Kitchen Essentials	12.00	3	36.00	No
2	1	2023-04-01	344	Veg Mango Ginger 100g	Fruits & Vegetables	16.00	3	48.00	No
3	1	2023-04-01	541	Whisper Ultra 3 Smart Layer XL 45s 1 No	Feminine CareÂ	569.05	3	1707.15	No
4	1	2023-04-01	551	Veet Cold Wax Strips Sensitive Skin 8 Strips	Feminine CareÂ	117.60	1	117.60	No
5	1	2023-04-01	417	Gopuram Kasthuri Turmeric Powder 40 gm	Household Essentials	21.85	1	21.85	No
6	1	2023-04-01	139	Maaretti Bruschette Roasted Garlic Bread 70 g	Instant Foods	94.05	2	188.10	No
7	1	2023-04-01	331	Longan Fruit 500g	Fruits & Vegetables	156.50	1	156.50	No
8	1	2023-04-01	276	Banana Yelakki 500g	Fruits & Vegetables	36.50	2	73.00	No
9	1	2023-04-01	506	Savlon Antiseptic Liquid 50 ml	Feminine CareÂ	26.50	3	79.50	No
10	1	2023-04-01	214	Colgate Cibaca Toothbrush 1 pc	Oral Care	14.25	2	28.50	No
11	1	2023-04-01	235	Dabur Red Paste Family Value Pack 2Â—175 g	Oral Care	266.94	1	266.94	No
12	1	2023-04-01	240	Tomato Country 1Kg	Fruits & Vegetables	71.50	2	143.00	No

Previewing first 50 entries.

Figure 4. Data Preview in R-Studio BillProduct

2.1.3 HANDLING MISSING VALUES AND DATA VALIDATION

Missing values appeared mainly in fields such as ProductRate, Number_of_Qty, Product_Category and occasionally Total_Amount. Rather than applying statistical imputation, the approach followed a business validation logic. If a transaction had quantity recorded but selling price was missing or zero, the line item was excluded from revenue-based indicators such as Profitability Comparison, Combo Revenue Impact and Price Sensitivity analysis.

Cross-checks were carried out between Number_of_Qty * ProductRate at item level and Total_Amount at bill level to identify obvious inconsistencies. Where mismatches were very large or where the combo flag was unclear, such records were removed from analytical tables used for KPI computation. This conservative handling protects the integrity of indicators like Basket Size, Revenue Share and Inventory Risk, and keeps interpretations aligned with realistic supermarket operations.

2.1.4 FEATURE TRANSFORMATION FOR COMBO ANALYTICS

Several derived variables were created using mutate(), group_by() and summarise() to support combo-focused business insights. In the product-level table, a price_band variable was constructed using cut() to classify selling prices into bands 0–50, 51–100, 101–200, 201–500 and 500+. This supports price sensitivity and combo distribution analysis across price ranges. At bill level, the total number of products per bill (Number_of_Products) and total revenue per bill were used to compute average basket size and average bill value for combo and non-combo segments.

Aggregate product-level measures such as combo_qty, noncombo_qty, combo_share_within_product and combo_share_within_all_combo were created to quantify how much of each product's movement comes through combos. Additional transformations produced product_speed for slow-moving items, a combo_dependency_pct indicator for discount dependency and a YearMonth variable for monthly combo conversion analysis. These engineered features do not alter the original business records. Instead, they create an analytical layer that reveals patterns in combo usage, pricing response, inventory risk and long-term purchase behaviour.

2.2 PLANNING AND DESIGN

The Gantt chart for this study maps the analytical workflow across clearly sequenced phases, starting from raw data handling and ending with modelling and reporting. In the first phase, Data Import and Structuring, supermarket billing and product files are loaded into R, variables are standardised and initial integrity checks are performed. The second phase, Data Cleaning and Preparation, covers missing value treatment, duplicate removal, category corrections and creation of the master bill–product dataset. The third phase, Exploratory Data Analysis (EDA) and KPI Definition, focuses on summary statistics, price-band profiling, combo vs non-combo comparisons and finalisation of key performance indicators such as combo conversion rate, revenue share, price-band mix and category contribution. The fourth phase, Feature Engineering and Advanced Metrics, includes construction of derived variables like price bands, combo dependency, inventory risk indicators and product-level contribution measures. The fifth phase, Modelling and Pattern Discovery, covers ARIMA and Prophet forecasting, k-means clustering for bill segmentation and Apriori-based market basket rule mining. The final phase, Visualisation, Interpretation and Report Compilation, is dedicated to building R-based dashboards and plots, interpreting results in business language and integrating all outputs into the final chapters. Each phase in the Gantt chart is shown with overlapping time bars where necessary, highlighting dependencies between data preparation, metric design, modelling and business interpretation.

Phase 1: Data Import & Structuring

This phase involves collecting all raw supermarket datasets, including bill-level files, product-wise line items, category lists and pricing tables. The data is imported into R using libraries such as readr, readxl or data.table. Initial structuring is done by converting date fields, checking

variable formats and merging datasets into a consistent relational structure. This phase ensures that the analytical foundation is technically sound before deeper processing begins.

Phase 2: Data Cleaning & Pre-Processing

During this stage, major data hygiene tasks are performed. Missing values are addressed, duplicate entries are removed, product name inconsistencies are corrected and category-level errors are fixed. Derived variables such as bill-wise totals, product-level rates and combo flags are validated. Outlier checks and normalisation procedures help stabilise the dataset. By the end of this phase, a clean analytical dataset is ready for exploration.

Phase 3: Exploratory Data Analysis (EDA) & KPI Development

This phase focuses on understanding patterns within the data through summary statistics, visualisations and structural profiling. Key performance indicators (KPIs) such as combo conversion rate, average bill value, revenue share, price-band distribution and category-wise contributions are formulated. EDA also reveals behavioural patterns such as product affinity, price sensitivity and monthly demand variations. These insights shape the direction of the advanced modelling phases.

Phase 4: Feature Engineering & Metric Construction

New analytical variables and metrics are created during this phase. Price bands, combo dependency indexes, inventory risk indicators, product-level contribution percentages and frequently paired product features are engineered. These variables help quantify subtle behavioural trends and provide deeper inputs to forecasting, clustering and market-basket models. This phase translates business behaviour into measurable analytical factors.

Phase 5: Forecasting & Predictive Modelling

This phase applies advanced models such as ARIMA and Prophet to forecast monthly combo revenue and identify seasonal patterns. K-means clustering is used to segment customers based on basket size, spend value and combo adoption. Apriori-based Market Basket Analysis is performed to uncover product affinities suitable for combo design and cross-sell strategies. This phase generates predictive insights that support operational decisions.

Phase 6: Visualisation, Interpretation & Report Compilation

In the final phase, all analytical results are visualised using ggplot2 and interpreted in clear business language. Dashboards, charts and tables are organised into report-ready formats. Technical findings are translated into managerial insights, business recommendations and strategic implications for supermarket operations. Final documentation is completed for inclusion in the report.

2.3 EXPLORATORY DATA ANALYTICS

Exploratory Data Analytics (EDA) was applied to convert raw billing transactions into structured insights about combo performance. The focus is descriptive rather than predictive, concentrating on how customers actually respond to combo offers in terms of bill counts, basket size, value, product mix, seasonality and category behaviour. Univariate analysis addresses individual variables such as combo status, category and price band distribution. Bivariate analysis compares combo and non-combo bills on basket size and bill value and identifies frequently paired product combinations. Time-based analysis examines monthly combo conversion trends, while category contribution analysis links combo behaviour with slow-moving items and discount dependency. EDA therefore forms the bridge between raw data and the key performance indicators that guide combo design and stock planning.

2.3.1 UNIVARIATE INSIGHTS

Univariate results show an almost even split between combo and non-combo bills. Out of 27,292 bills, 13,740 bills have at least one combo and 13,552 bills do not, giving a Combo Conversion Rate of about 50.3 percent. This metric, plotted as a simple bar chart, indicates that every second bill carries a combo, which confirms that customers are familiar with the scheme and engage with it regularly.

The distribution of product categories at line-item level reveals that Fruits and Vegetables, Household Essentials and Feminine Care together account for a major share of item movement, while categories like Baby Needs and Rice Products contribute smaller volumes. Price-band analysis shows that around 42 percent of items fall in the 0–50 range, nearly 24 percent in 51–100 and close to 19 percent in 101–200, with higher bands contributing progressively less. This confirms that the supermarket is dominated by low and mid-priced items, which sets an important context for interpreting combo design and discount depth.

2.3.2 BIVARIATE INSIGHTS

Bivariate analysis involves comparing two variables to understand the effect of combos on basket characteristics and monetary performance. For example, comparing the number of items purchased in combo and non-combo bills helps compute the Basket Size Lift Model, which determines whether bundling strategies are truly encouraging customers to buy more or simply replacing individual purchases. An increase in basket size indicates successful cross-selling, while a negligible difference suggests that combo offers attract buyers only for discounted single-product value rather than encouraging bulk purchase.

Additionally, comparison of bill values between combo and non-combo transactions supports the Profitability Comparison Model, which examines whether combos enhance revenue or simply increase product movement without profit gain. Observing quantity per bill also contributes to the evaluation of inventory usage. Meanwhile, identifying frequently paired products using item-to-item comparisons directly supports the Product Pairing Model, which reveals which combinations are commonly chosen by customers and which category pairs show greater lift. Such insights highlight opportunities for data-driven combo structuring, rather than random discount assignments.

2.3.3 TIME-BASED ANALYSIS

Time-based inspection identifies seasonal patterns of customer response to combo offers. By analysing combo conversion month-wise, supermarkets can identify peak months where combo adoption increases, supporting the Seasonal Combo Effectiveness Model. Spikes may appear before festive periods such as Diwali, Pongal, or New Year, or near month-end when household stock replenishment typically occurs. This variation signifies the importance of strategic timing in promotional planning, where offering combos during high-response periods yields better conversion and revenue.

Time-based insights also connect to the Revenue Impact of Combo Purchases, since observing periods where combo sales contribute a larger share of revenue provides input for promotional budgeting. If revenue contribution from combos peaks seasonally, stores may optimise stock allocation and discount strategies during high-purchase months. These analytics highlight that combo offers do not perform uniformly throughout the year and require seasonally aligned decision-making.

2.3.4 CATEGORY CONTRIBUTION INSIGHTS

Category contribution analysis evaluates how different product categories perform when included in combo offers. Some categories may show high lift, indicating strong customer acceptance when bundled, while others may exhibit weak response despite frequent discounting. This directly links to the Category-Wise Combo Lift Index, providing clarity on where stores should focus or reduce bundling efforts. Categories repeatedly used in combos but still showing weak revenue or quantity performance may reflect mismatch between customer needs and bundling strategy.

This section also identifies slow-moving products repeatedly pushed into combos, which directly supports the Combo Inventory Risk Model. If such products continue to underperform even when bundled, it signals ineffective clearance strategy. Meanwhile, products purchased only through combos cater to Discount Dependency Analysis, where more than 60% dependency indicates that customers do not value them without discounts. These insights collectively help stores decide which products to retain, rotate, or remove from discount strategies, aligning combo offers with market demand rather than assumption-based planning.

2.4 MODEL / INSIGHT SECTION

Beyond descriptive EDA, the project can optionally extend into more advanced analytical models. Forecasting models such as ARIMA or Prophet can be used to project category-wise or combo-wise sales for future months. Customer segmentation through k-means clustering can group bills or customers based on basket size, spending level and combo usage intensity. Market basket rules generated using the Apriori algorithm can formalise association rules for product pairing, which supports automated combo suggestion. These models would deepen the current descriptive findings and provide a pathway towards semi-automated decision support for supermarket management.

2.5 ACCURACY METRICS AND BUSINESS INTERPRETATION

Where predictive or segmentation models are introduced, standard metrics such as Mean Absolute Percentage Error for forecasts or silhouette scores for clustering can be reported. However, for this project the primary emphasis is on business interpretability rather than algorithmic accuracy. The value of the analysis lies in clear metrics such as Combo Conversion Rate, Basket Size comparison, Revenue Share, Inventory Risk flags and Discount Dependency percentages, all of which can be directly understood by store managers. Each

visual and table is interpreted in plain language, explaining whether combos genuinely improve sales mix, which products carry risk and how seasonal timing influences performance.

2.6 SCREENSHOTS, CODE AND GRAPHS WITH BUSINESS MEANING

All important R code blocks, graphs and summary tables are documented with screenshots. Each output is accompanied by a short explanation of what it means for the supermarket. For example, bar charts for combo vs non-combo basket size, revenue share charts, price band distributions and monthly conversion line plots are linked to decisions on stock planning, promotion intensity and product pairing. This structure ensures that technical work in R is transparently connected to practical retail decisions.

2.6.1 COMBO VS NON-COMBO BASKET SIZE

The basket size analysis compares the average number of products purchased in bills with combo offers and those without combos. The computed values show that both combo and non-combo bills have almost identical average basket sizes at around 16.5 items per bill, with similar variability (standard deviation ≈ 6.9). This indicates that combo offers are not significantly influencing customers to increase the number of items purchased. Instead, customers purchase almost the same number of products irrespective of whether a combo is chosen, suggesting that combos attract buyers but do not necessarily drive higher basket volume or additional cross-selling. Based on this result, supermarket combo strategies may require better product pairing or stronger promotional structures to successfully lift basket size.

Algorithm

Group Bills by Combo Selection

The dataset is first split into two groups based on whether a bill opted for a combo (Combo_Opted = "Yes" or "No").

1. **Calculate Bill Counts**

   For each group, the total number of bills is counted to understand transaction volume under each condition.

2. **Compute Average Basket Size**

   The mean (average) number of items (Number_of_Products) in each bill is calculated within each group to measure the basket size trend.

3. **Measure Variation (Standard Deviation)**

   The standard deviation is computed to understand how much basket sizes vary among customers within each group.

4. **Visualize for Comparison**

   A bar chart is created to compare average basket sizes, and labels are added for clarity. Different colors are used for combo vs. non-combo to highlight differences visually.

Coding

```
basket_size_summary <- bill_main %>%
+     group_by(Combo_Opted) %>%
+     summarise(
+       count_bills       = n(),
+       avg_items_per_bill = mean(Number_of_Products, na.rm = TRUE),
+       sd_items_per_bill  = sd(Number_of_Products, na.rm = TRUE)
+     )
>
> basket_size_summary
> ggplot(basket_size_summary, aes(x = Combo_Opted, y = avg_items_per_bill, fill = Combo
_Opted)) +
+     geom_col(width = 0.5) +
+     geom_text(aes(label = round(avg_items_per_bill, 2)),
+               vjust = -0.5, size = 4, fontface = "bold") +
+     labs(
+       title = "Average Basket Size: Combo vs Non-Combo Bills",
+       x = "Combo Opted",
+       y = "Average Number of Products"
+     ) +
+     scale_fill_manual(values = c("Yes" = "#1f77b4", "No" = "#ff7f0e")) +
+     theme_minimal() +
+     theme(
+       plot.title = element_text(face = "bold", hjust = 0.5),
+       legend.position = "none"
+     )
```

Output :

```
# A tibble: 2 × 4
  Combo_Opted count_bills avg_items_per_bill sd_items_per_bill
  <chr>           <int>              <dbl>             <dbl>
1 No              13552               16.5              6.91
2 Yes             13740               16.5              6.93
```

INFERENCE :

The average basket size for both combo and non-combo purchases is almost the same, at roughly 17 items per bill. This shows that combo offers are not encouraging customers to add more products, despite appearing attractive. Therefore, combos are driving purchase decisions through pricing alone and not delivering additional cross-selling benefits.

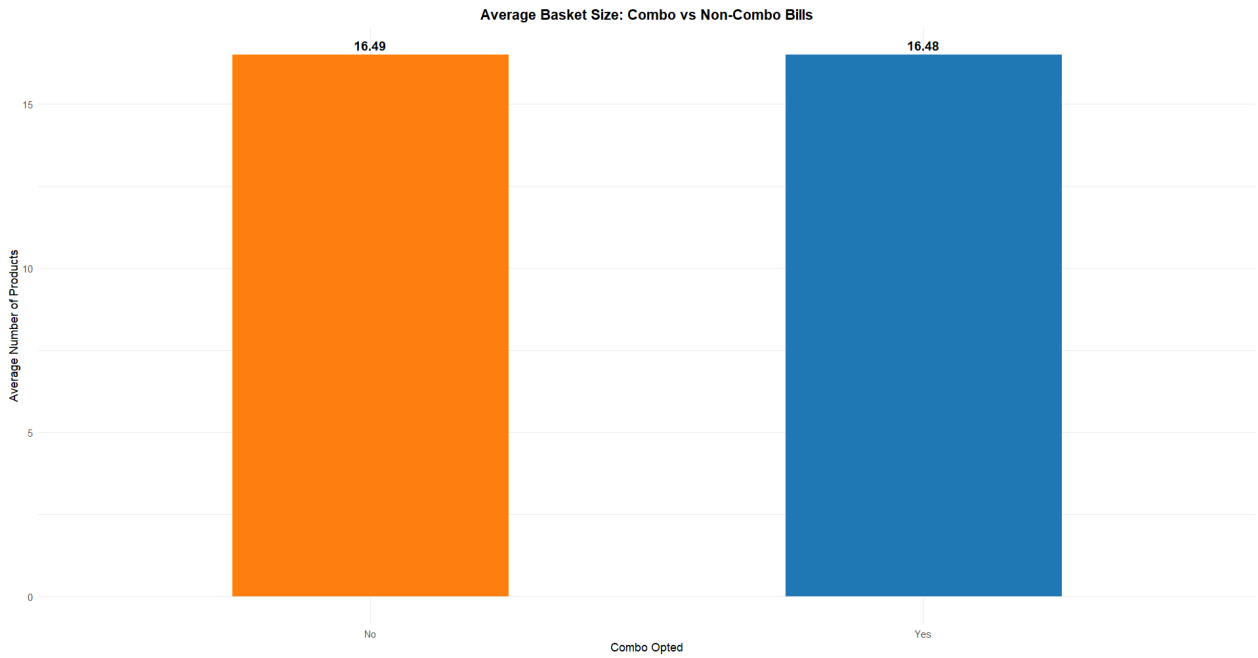

Figure 7. Average Basket Size: Combo vs Non-Combo Bills

2.6.2 COMBO VS NON-COMBO BILL VALUE

The comparison of bill values shows that the average spending is almost identical between combo and non-combo bills. Even though combo bills are expected to increase spending through bundled pricing, customers spend approximately the same amount with or without combos. This indicates that combo offers may be acting as substitutes rather than add-ons, attracting buyers without significantly increasing bill value.

Algorithm

1. Separate bills into two groups based on whether a combo was purchased (Yes or No).
2. Count how many bills are in each group to know their total volume.
3. Calculate the average bill value for combo and non-combo bills.
4. Calculate the standard deviation to understand variation in bill spending.
5. Compare the averages to check whether combos lead to higher bill value.

Coding

```
> bill_value_summary <- bill_main %>%
+     group_by(Combo_Opted) %>%
+     summarise(
+       count_bills     = n(),
+       avg_bill_value  = mean(Total_Amount, na.rm = TRUE),
+       sd_bill_value   = sd(Total_Amount, na.rm = TRUE)
+     )
>
> bill_value_summary

> ggplot(bill_value_summary, aes(x = Combo_Opted, y = avg_bill_value, fill = Combo_Opt
ed)) +
+     geom_col(width = 0.5) +
+     geom_text(aes(label = round(avg_bill_value, 2)),
+               vjust = -0.5, size = 4, fontface = "bold") +
+     labs(
+       title = "Average Bill Value: Combo vs Non-Combo Bills",
+       x = "Combo Opted",
+       y = "Average Bill Value (₹)"
+     ) +
+     scale_fill_manual(values = c("Yes" = "#2ca02c", "No" = "#d62728")) +
+     theme_minimal() +
+     theme(
+       plot.title = element_text(face = "bold", hjust = 0.5),
+       legend.position = "none"
+     )
```

Output :

```
# A tibble: 2 × 4
  Combo_Opted count_bills avg_bill_value sd_bill_value
  <chr>           <int>         <dbl>         <dbl>
1 No              13552         2879.         1433.
2 Yes             13740         2877.         1435.
```

Inference :

Although the average bill value for combo bills (₹2877) is nearly equal to non-combo bills (₹2879), the result is encouraging because combo purchases achieve the same revenue level without reducing spending, even when discounts are provided. This shows that customers do not restrict their spending while opting for combos, and the bundled pricing helps maintain bill value while promoting higher product movement. Therefore, combo offers are a commercially viable strategy to sustain revenue and stimulate bundled sales without lowering the average earning per bill.

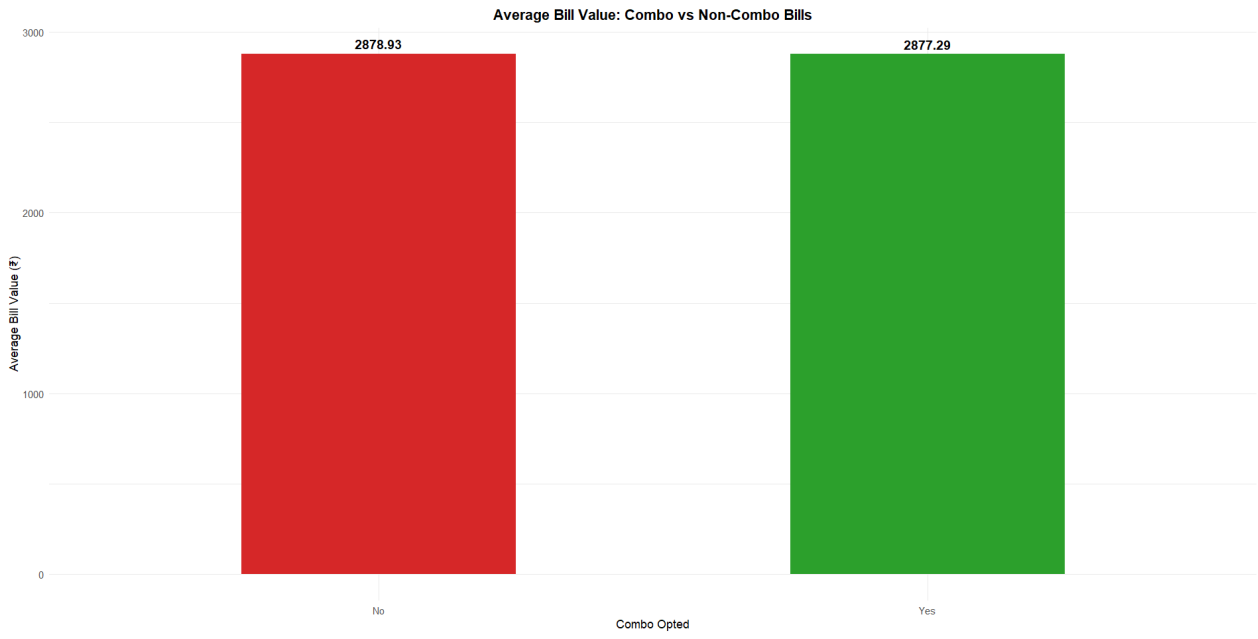

Figure 8. Average Bill Value: Combo vs Non-Combo Bills

2.6.3 FREQUENTLY PAIRED PRODUCTS

The analysis of frequently paired products reveals the top combinations of items that customers most often purchase together, offering valuable insights into natural buying patterns and cross-selling potential. By counting the frequency of each product pair and identifying the top 50 combinations, the results highlight which items have strong complementary relationships in real purchasing behaviour. These high-frequency pairs can be strategically used for combo creation, targeted promotions, product placement, and personalized recommendations. Overall, this finding strengthens merchandising decisions by showing which product combinations consistently drive customer preference and increase basket value.

Coding :

```
View(product_pairs)
> library(dplyr)
>
> top_50_pairs <- product_pairs %>%
+     count(Product_A, Product_B, sort = TRUE) %>%   # count how many times each pair a
ppears
+     head(50)                                  # keep only top 50
>
> top_50_pairs
```

Output :

```
> bill_main    <- BillMain
> bill_product <- BillProduct
> head(bill_main)
# A tibble: 6 × 5
  BillNo BillDate            Number_of_Products Total_Amount Combo_Opted
   <dbl> <dttm>                           <dbl>        <dbl> <chr>
1      1 2023-04-01 00:00:00                 20         2780 No
2      2 2023-04-01 00:00:00                 21         2331 Yes
3      3 2023-04-01 00:00:00                 10         1220 No
4      4 2023-04-01 00:00:00                 10         2290 No
5      5 2023-04-01 00:00:00                 22         5236 Yes
6      6 2023-04-01 00:00:00                 18         4068 Yes
> head(bill_product)
# A tibble: 6 × 10
  TransNo BillNo BillDate            PCode Product_Name      Product_Category ProductRate
    <dbl>  <dbl> <dttm>              <dbl> <chr>             <chr>                  <dbl>
1       1      1 2023-04-01 00:00:00    24 Stone Salt 1Kg    Kitchen Essenti…        12
2       2      1 2023-04-01 00:00:00   344 Veg Mango Ginger… Fruits & Vegeta…        16
3       3      1 2023-04-01 00:00:00   541 Whisper Ultra 3 … Feminine CareÂ         569.
4       4      1 2023-04-01 00:00:00   551 Veet Cold Wax St… Feminine CareÂ         118.
5       5      1 2023-04-01 00:00:00   417 Gopuram Kasthuri… Household Essen…        21.8
6       6      1 2023-04-01 00:00:00   139 Maaretti Brusche… Instant Foods           94.0
# ℹ 3 more variables: Number_of_Qty <dbl>, Total_Amount <dbl>, Combo_Opted <chr>
>
```

2.6.4 TOTAL NUMBER OF BILLS WITH AND WITHOUT COMBO PURCHASES (COMBO CONVERSION RATE)

The analysis of total bills with and without combo purchases helps understand how effectively customers are choosing bundled offers during their transactions. By counting the number of bills under each category, the results show that 13,740 bills included combo purchases while 13,552 bills did not. To measure this behaviour more precisely, the combo conversion rate was calculated by dividing the number of combo bills by the total bill count. This metric indicates the proportion of customers who preferred value-added combo options over individual items, offering a clear view of customer buying patterns and the effectiveness of combo strategies.

Coding

```
> # Count of bills by combo status
> bill_combo_counts <- bill_main %>%
+     count(Combo_Opted)
>
> bill_combo_counts
# A tibble: 2 × 2
  Combo_Opted     n
  <chr>       <int>
1 No          13552
2 Yes         13740
>
```

```
> # Compute combo conversion rate
> combo_conversion <- bill_main %>%
+     summarise(
+         total_bills  = n(),
+         combo_bills  = sum(Combo_Opted == "Yes", na.rm = TRUE),
+         noncombo_bills = sum(Combo_Opted == "No", na.rm = TRUE),
+         combo_conversion_rate = combo_bills / total_bills * 100
+     )
> combo_conversion
ggplot(bill_main, aes(x = Combo_Opted)) +
+     geom_bar() +
+     labs(
+         title = "Number of Bills With and Without Combo",
+         x = "Combo Opted",
+         y = "Number of Bills"
+     )
> library(ggplot2)
> library(dplyr)
>
> # Count bills
> combo_plot_data <- bill_main %>%
+     count(Combo_Opted)
>
> # Colorful bar plot with labels
> ggplot(combo_plot_data, aes(x = Combo_Opted, y = n, fill = Combo_Opted)) +
+     geom_col(width = 0.6) +
+     geom_text(aes(label = n), vjust = -0.5, size = 5, fontface = "bold") +
+     scale_fill_manual(values = c("Yes" = "#1fa8e7", "No" = "#ff7f0e")) +
+     labs(
+         title = "Bills With vs Without Combo Purchase",
+         x = "Combo Opted",
+         y = "Number of Bills",
+         fill = "Combo Selection"
+     ) +
+     theme_minimal() +
+     theme(
+         plot.title = element_text(face="bold", size = 14, hjust = 0.5),
+         axis.text = element_text(size = 12, face = "bold"),
+         axis.title = element_text(size = 13, face = "bold"),
+         legend.position = "none"
+     )
```

Output:

```
# A tibble: 1 × 4
  total_bills combo_bills noncombo_bills combo_conversion_rate
        <int>       <int>          <int>                 <dbl>
1       27292       13740          13552                  50.3
```

INFERENCE : The conversion results show a strong customer inclination towards combo purchases, as bills with combos slightly exceed non-combo bills. This indicates that combo offers are well-accepted and are successfully motivating customers to choose bundled products, suggesting a positive impact on sales volume and overall bill value.

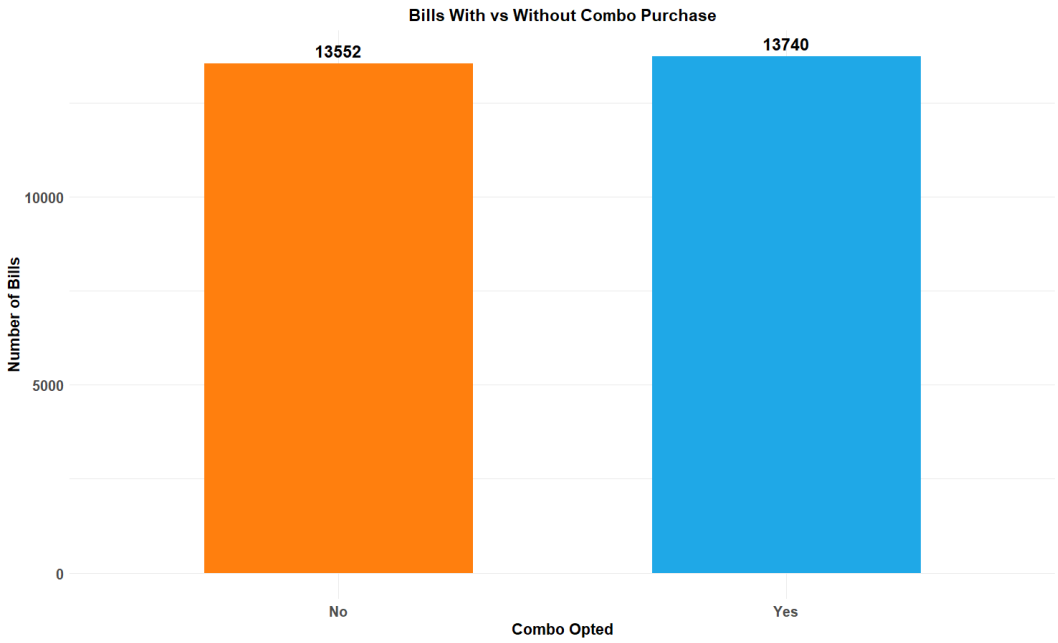

Figure 9. Bills with versus Without Combo Purchase

2.6.5 DISTRIBUTION OF PRODUCT CATEGORIES

The distribution of product categories highlights how customer purchasing behaviour varies across different segments, offering a clear picture of which categories dominate overall sales volume. The category-wise count shows that Fruits & Vegetables lead with the highest number of line items, followed closely by Household Essentials and Feminine Care, each contributing significantly to total transactions. Mid-level categories such as Staples, Kitchen Essentials, and Breakfast Cereals also form a stable share of purchases, while niche segments like Baby Needs and Rice Products contribute smaller proportions. By converting these counts into percentages, the analysis provides a structured understanding of category presence, enabling the business to identify which product groups drive the most frequent purchases and which segments need focused attention for growth, stocking, and promotions.

Coding

```
> # Category-wise count of line items
> category_counts <- bill_product %>%
+     count(Product_Category, sort = TRUE) %>%
+     mutate(percentage = round(n / sum(n) * 100, 2))
```

```
>
> category_counts


> library(ggplot2)
>
> # Count categories again (just to be safe)
> category_counts <- bill_product %>%
+     count(Product_Category, sort = TRUE)
>
> ggplot(category_counts, aes(x = reorder(Product_Category, -n), y = n, fill = Product_Categ
ory)) +
+     geom_bar(stat = "identity") +
+     geom_text(aes(label = n),
+               position = position_stack(vjust = 0.5),
+               color = "white",
+               fontface = "bold",
+               size = 4) +
+     labs(
+       title = "Distribution of Product Categories",
+       x = "Product Category",
+       y = "Number of Line Items"
+     ) +
+     theme_minimal(base_size = 13) +
+     theme(
+       plot.background = element_rect(fill = "#1a1a1a", color = NA),
+       panel.background = element_rect(fill = "#1a1a1a"),
+       panel.grid.major = element_line(color = "#444444"),
+       panel.grid.minor = element_line(color = "#333333"),
+       plot.title = element_text(size = 14, face = "bold", hjust = 0.5, color = "white"),
+       axis.text.x = element_text(angle = 45, hjust = 1, size = 11, face = "bold", color = "whit
e"),
+       axis.text.y = element_text(size = 11, face = "bold", color = "white"),
+       axis.title = element_text(size = 12, face = "bold", color = "white"),
+       legend.position = "none"
+     ) +
+     scale_fill_brewer(palette = "Dark2")
```

Output :

```
Product_Category                n percentage
  <chr>                     <int>      <dbl>
 1 Fruits & Vegetables        97521      21.7
 2 Household Essentials       81566      18.1
 3 Feminine CareÂ             81547      18.1
 4 Staples                 26034       5.79
 5 Kitchen Essentials         25826      5.74
 6 Breakfast Cereals          20313      4.51
 7 Masala & Spices            19470      4.33
 8 Oral Care                 19400       4.31
```

9	Instant Foods	18782	4.17
10	Bread Spreads, Sauces & Mayonnaise	18019	4
11	Appalam & Papads	18008	4
12	Rice & Rice Products	13001	2.89
13	Baby Needs	10491	2.33

INFERENCE : The distribution clearly shows that customer demand is heavily concentrated in a few major categories. Fruits & Vegetables lead with 97,521 line items (21.7%), indicating that fresh produce drives the highest footfall and repeat purchases. This is followed by Household Essentials with 81,566 items (18.1%) and Feminine Care with 81,547 items (18.1%), reflecting strong demand for daily-use and personal-care products. Mid-tier categories such as Staples (26,034; 5.79%) and Kitchen Essentials (25,826; 5.74%) show stable, moderate buying patterns. Lower-share categories like Rice & Rice Products (13,001; 2.89%) and Baby Needs (10,491; 2.33%) contribute smaller portions but still represent steady, essential-use segments. Overall, the values indicate that essential, high-frequency categories dominate purchases, while mid and low-volume groups offer opportunities for targeted growth and promotional strategies.

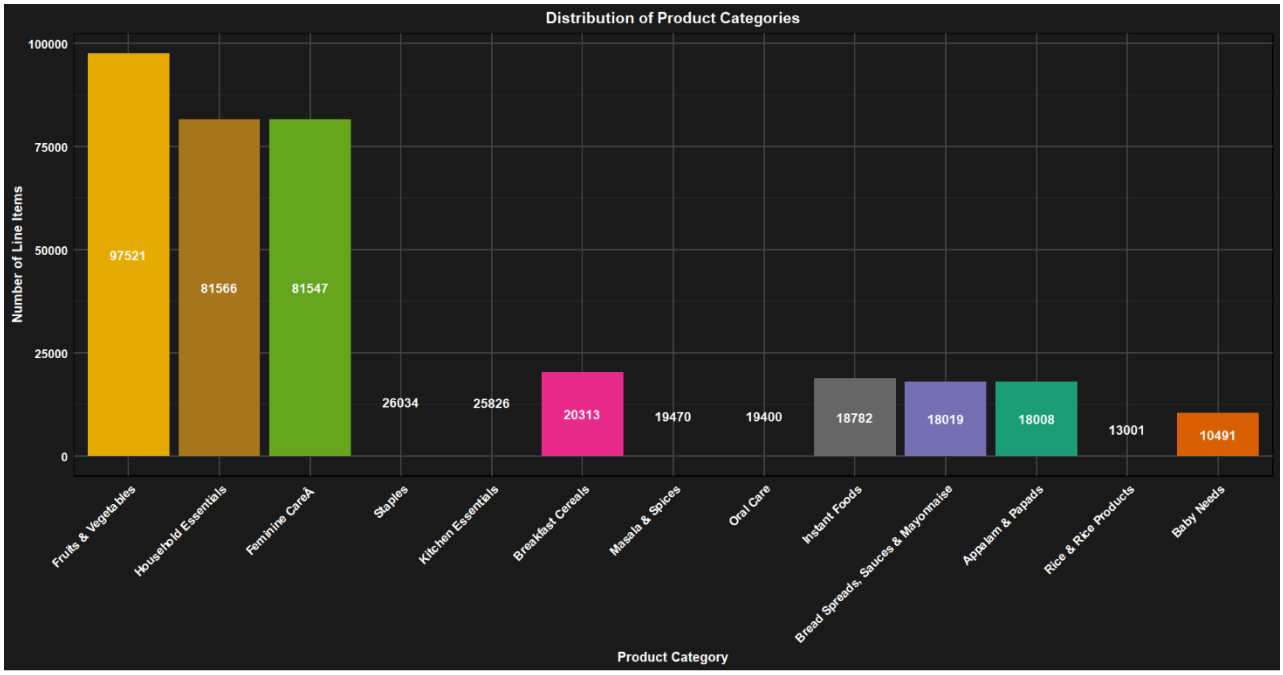

Figure 10. Distribution of Product Categories

2.6.6 PRICE SPREAD AND COMMONLY PURCHASED PRICE RANGES

The analysis of price spread and commonly purchased price ranges aims to understand how customers distribute their spending across various product price levels. By summarizing the minimum, median, and maximum product rates, and then grouping items into defined price bands, this section provides a clear picture of where the bulk of customer purchases occur. Price segmentation into ranges such as 0–50, 51–100, 101–200, 201–500, and 500+ helps identify affordability patterns, value preferences, and the typical spending behaviour across the store's assortment. This breakdown supports pricing decisions, promotional planning, and inventory management by highlighting which price brackets attract the highest customer engagement.

Coding

```
> # Basic summary of product prices
> summary(bill_product$ProductRate)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    5.0    31.5    66.5   115.2   139.7  1599.0
>
> # Create price bands
> bill_product <- bill_product %>%
+     mutate(
+       price_band = cut(
+         ProductRate,
+         breaks = c(0, 50, 100, 200, 500, Inf),
+         labels = c("0-50", "51-100", "101-200", "201-500", "500+"),
+         right = TRUE
+       )
+     )
>
> # Count by price band
> price_band_counts <- bill_product %>%
+     count(price_band, sort = TRUE) %>%
+     mutate(percentage = round(n / sum(n) * 100, 2))
>
> price_band_counts
> ggplot(bill_product, aes(x = ProductRate)) +
+     geom_histogram(binwidth = 20, boundary = 0, closed = "right") +
+     labs(
+       title = "Price Spread of Products",
+       x = "Product Rate",
+       y = "Frequency"
+     )
```

Output :

```
# A tibble: 5 × 3
  price_band      n percentage
```

	<fct>	<int>	<dbl>
1	0-50	188860	42.0
2	51-100	107362	23.9
3	101-200	86511	19.2
4	201-500	57477	12.8
5	500+	9768	2.17

INFERENCE : The results show that a majority of purchases fall within lower and mid-level price ranges, indicating strong consumer preference for affordable and value-driven products. The 0–50 range dominates with 188,860 line items (42%), followed by 51–100 with 107,362 items (23.9%) and 101–200 with 86,511 items (19.2%), confirming that about 85% of all pur chases occur below ₹200. Higher-priced ranges such as 201–500 (12.8%) and 500+ (2.17%) contribute smaller but steady volumes. Overall, the inference suggests that customers prefer e conomical products, making budget-friendly pricing and attractive combo offers essential stre ngths for driving sales and maintaining high purchase frequency.

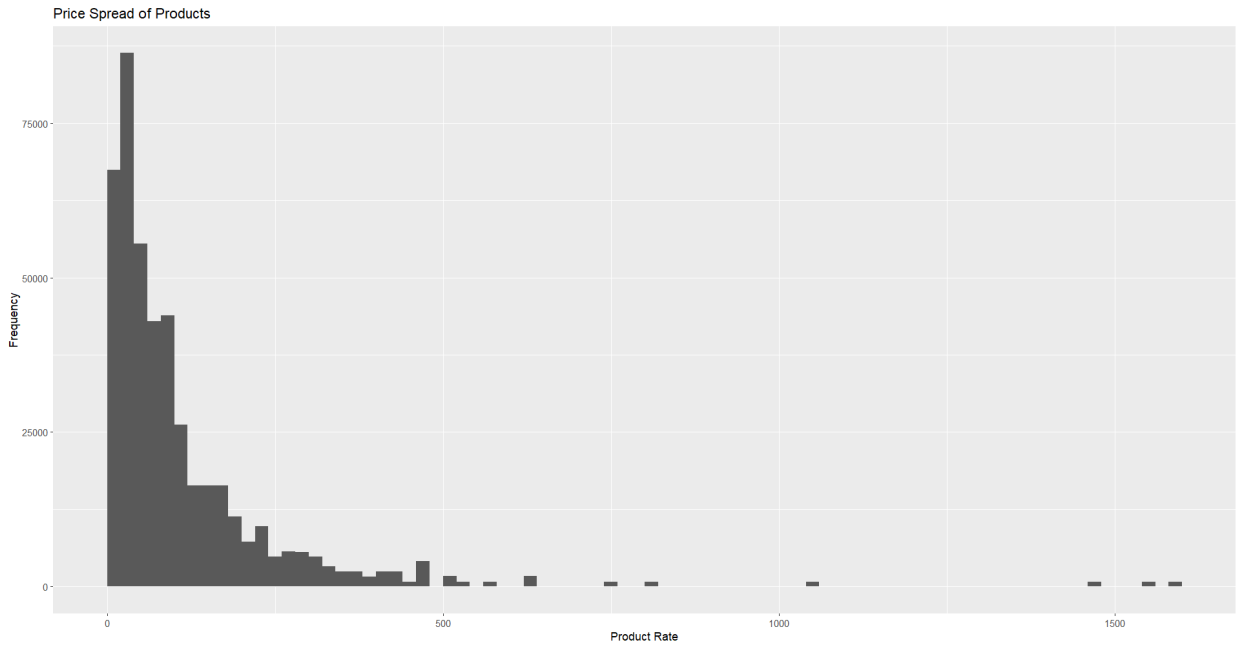

Figure 11. Price Spread of Products

Commonly Purchased Price Ranges

Coding

```
> # Calculate percentage
> price_band_counts <- price_band_counts %>%
+     mutate(percent = round(n / sum(n) * 100, 2))
>
> # Plot
```

```
> ggplot(price_band_counts, aes(x = price_band, y = percent, fill = price_band)) +
+   geom_bar(stat = "identity") +
+   geom_text(aes(label = paste0(percent, "%")),
+             vjust = -0.5, size = 4, fontface = "bold", color = "black") +
+   labs(
+     title = "Commonly Purchased Price Ranges (Percentage)",
+     x = "Price Band (₹)",
+     y = "Percentage of Line Items"
+   ) +
+   theme_minimal() +
+   theme(
+     plot.title = element_text(face = "bold", hjust = 0.5),
+     axis.text.x = element_text(size = 11, face = "bold"),
+     legend.position = "none"
+   ) +
+   scale_fill_brewer(palette = "Dark2")
```

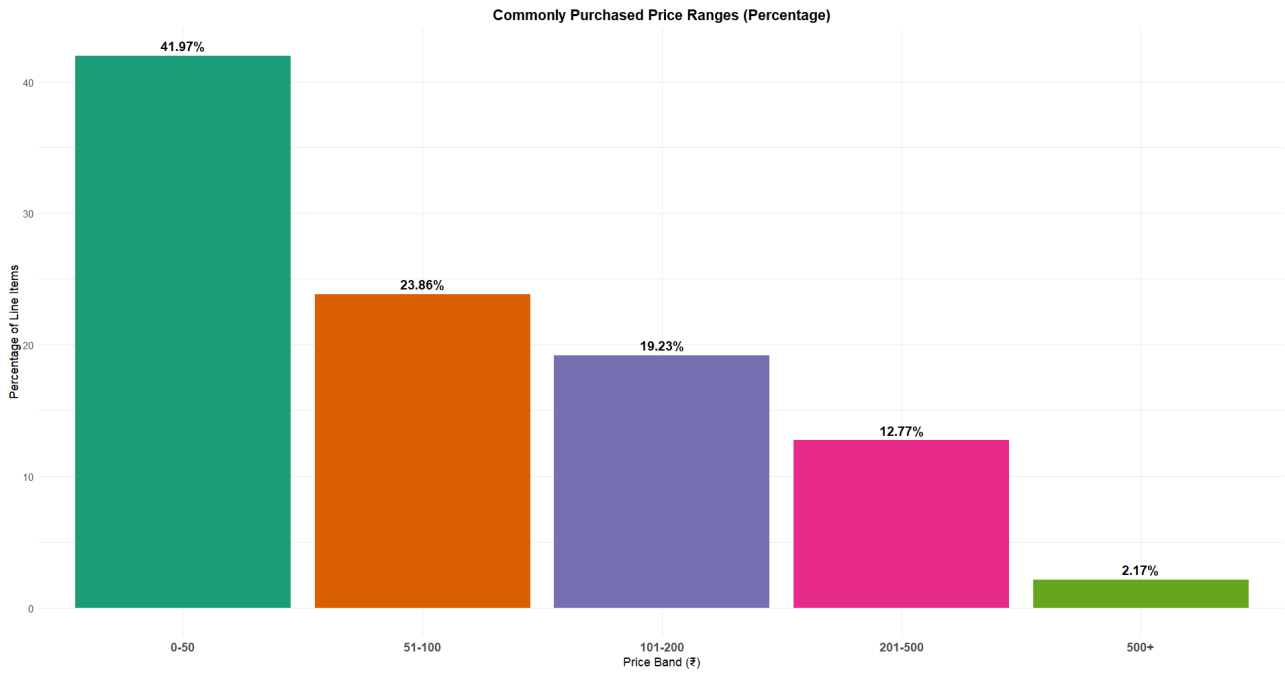

Figure 12. Commonly Purchased Price Ranges

2.6.7 TOP 5 MOST FREQUENT PRODUCT PAIRS INSIDE THE CATEGORY

This section identifies the top five most frequently purchased product pairs within each category, providing deeper insight into how customers combine items during their shopping. By analyzing unique product combinations within the same bill, the model highlights natural associations between products that belong to the same category. These patterns help understand customer preferences, reveal strong complementary relationships, and support

strategic decisions such as category-level bundling, shelf placement, promotional offers, and combo creation.

Coding

```
> # Keep only BillNo and Product_Name
> pairs_data <- bill_product %>%
+     select(BillNo, Product_Name) %>%
+     distinct()  # avoid duplicate same-product in same bill
>
> # Function to generate all unique pairs inside one bill
> get_pairs <- function(products) {
+     if(length(products) < 2) return(NULL)
+     t(combn(sort(unique(products)), 2)) %>% as.data.frame()
+ }
>
> library(purrr)
>
> product_pairs <- pairs_data %>%
+     group_by(BillNo) %>%
+     summarise(pairs = list(get_pairs(Product_Name)), .groups = "drop") %>%
+     tidyr::unnest(pairs, names_repair = "minimal")


> write.csv(top5_pairs_each_category,
+           "Top5_Product_Pairs_By_Category.csv",
+           row.names = FALSE)
>
> top50_pairs_each_category <- same_cat_pairs %>%
+   group_by(Category_A, Product_A, Product_B) %>%
+   summarise(pair_count = n(), .groups = "drop") %>%
+   group_by(Category_A) %>%
+   slice_max(order_by = pair_count, n = 5, with_ties = FALSE) %>%
+   arrange(Category_A, desc(pair_count))
>
> top5_pairs_each_category
```

Output :

Category_A	Product_A	Product_B	pair_count
Appalam & Papads	Ambika Appalam 325 gm	Popular Dinner Special Appalam 100 gm	41
Appalam & Papads	Popular Appalam 200 gm	Sago Cumin Papad 100 gm	38
Appalam & Papads	Bat Pappad 100 gm	Popular Masala Pappad 200 gm	37
Appalam & Papads	Cumin & Jeera Pappad 100 gm	Mothers Recipe Rice Papad Ajwain 75 gm	37

Appalam & Papads	Aachi Appalam 100 gm	Potato Papad 100 gm	36
Baby Needs	Nestle Cerelac Wheat Apple Cherry Stage 2 (8 Months+) 300 g	Nestle Lactogen Powder 3 (After 12 Months) 400 g	36
Baby Needs	Nestle Lactogen Powder 3 (After 12 Months) 400 g	Nestle Nan Pro 2 (After 6 Months) 400 g	36
Baby Needs	Nestle Cerelac 5 Grains & Fruits Stage 5 300 g	Nestle Cerelac Wheat Orange 300 g	35
Baby Needs	Nestle Cerelac 5 Grains & Fruits Stage 5 300 g	Nestle Cerelac Rice Vegetables Stage 2 (8 Months+) 300 g	34
Baby Needs	Nestle Cerelac Multigrain Dal Veg Stage 4 300 g	Nestle Cerelac Wheat Orange 300 g	34
Bread Spreads, Sauces & Mayonnaise	Dr. Oetker Funfood Peanut Butter Creamy Jar 220 g	Maaretti Bruschette Roasted Garlic Bread 70 g	64
Bread Spreads, Sauces & Mayonnaise	Funfoods Veg Mayonnaise Original 800 g	Maaretti Bruschette Roasted Garlic Bread 70 g	63
Bread Spreads, Sauces & Mayonnaise	Kissan Peanut Butter 100 g	Maaretti Bruschette Roasted Garlic Bread 70 g	59
Bread Spreads, Sauces & Mayonnaise	Maaretti Bruschette Roasted Garlic Bread 70 g	Normalife Strawberry Chia Healthy Fruit Spread 200 g	56
Bread Spreads, Sauces & Mayonnaise	24 Mantra Organic Creamy Peanut Butter 450 g	Maaretti Bruschette Roasted Garlic Bread 70 g	55
Breakfast Cereals	Kelloggs Chocos 110 g	Saffola Masala Oats Masala Coriander 39 g	40
Breakfast Cereals	Kelloggs Corn Flakes Original 100 g	Saffola Oats Pouch 500 g	40
Breakfast Cereals	Kelloggs Chocos Fills 180 g	Kelloggs Granola Almonds & Cranberries 150 g	39
Breakfast Cereals	Saffola Munchiez Ragi Chips Masala Twist 80 g	Slurrp Farms Millet Strawberry Stars & Moons Crunch 400 g	38
Breakfast Cereals	Kelloggs Granola Almonds & Cranberries 150 g	Kwality Breakfast Cereals Combo Pouch 6Ãƒâ€”20 g	36
Feminine Care	Veet Cold Wax Strips Sensitive Skin 8 Strips	Veet Hair Removal Cream Silk Fresh Sensitive 50 g	46
Feminine Care	Fix Shadow SPF 50 Cream 40 g	Stayfree Dry Max All Night Ultra Dry XL 18 Pads	45
Feminine Care	Savlon Antiseptic Liquid 50 ml	Vcare Hair Removal Spray Foam 50 ml	45
Feminine Care	Whisper Bindazzz Nights Koala Soft XXL + 5 Pads 1 No	Whisper Ultra Soft XL Sanitary Pads 6 Pads	44

Feminine Care	nua Zero Irritation Promise 10N	Stayfree Dry Max XXL 28 Pads	44
Fruits & Vegetables	Musk Melon 1.5Kg	Pomegranate Kabul 500g	46
Fruits & Vegetables	Cluster Beans 250g	Cucumber Country 500g	44
Fruits & Vegetables	Banana Karpooravalli 500g	Yam/Karunaikizhangu 500g	43
Fruits & Vegetables	Banana Morris 500g	Greens Palak 1 Bunch	43
Fruits & Vegetables	Brinjal Suphol 500g	Mahali Potato 1Kg	43
Household Essentials	Good Knight Gold Flash Liquid 45 ml	Odo Fresh Lavender Sanicubes 180 gm	49
Household Essentials	Good Knight Gold Flash Liquid 45 ml	Gopuram Turmeric Powder Tin 20 gm	48
Household Essentials	Bison Naphthalene Balls 100 gm	Om Santhi Pure Chandan Powder 50 gm	45
Household Essentials	Cycle Rhythm Orient Cones 22 gm	Lia Chandanam Room Freshener 160 ml	43
Household Essentials	Asian Globus Casserole 1N	Bison Naphthalene Balls 100 gm	42
Instant Foods	Maaretti Bruschette Roasted Garlic Bread 70 g	Snac Atac Lentil Twister 90 g	63
Instant Foods	Maaretti Bruschette Roasted Garlic Bread 70 g	McCain Chilli Garlic Potato Bites 200 g	60
Instant Foods	Dabur Hommade Tomato Puree 200 g	Maaretti Bruschette Roasted Garlic Bread 70 g	58
Instant Foods	Maaretti Bruschette Roasted Garlic Bread 70 g	Maretti Bruschette Chips Pizza Al Forno 70 g	58
Instant Foods	Maaretti Bruschette Roasted Garlic Bread 70 g	Maretti Bruschette Sour Cream & Onion 70 g	58
Kitchen Essentials	Fortune Double Roasted Suji Rawa 500g	Udhaiyam White Orid Dhall 1kg	42
Kitchen Essentials	Fortune Double Roasted Suji Rawa 500g	Virali Manjal 250g	39
Kitchen Essentials	BeroÃ¢â‚¬â„¢s Synthetic Vinegar 600ml	Indu Rock Salt 1Kg	38
Kitchen Essentials	Diamond Sugar (Kalkandu) 250g	Tata Stone Salt 1Kg	37
Kitchen Essentials	Panakarkandu Palm Candy 250g	Virali Manjal 250g	37
Masala & Spices	Aachi Chilli Powder 50 g	Sakthi Chilli Powder 50 g	45
Masala & Spices	Darling Idly Chilly Powder 50 g	MTR Chilli Powder 100 g	44
Masala & Spices	Eastern Asafoetida / Hing Powder 25 g	Eastern Turmeric Powder 100 g	43
Masala & Spices	Darling Turmeric Powder 100 g	Sakthi Turmeric Powder 50 g	40

Masala & Spices	MTR Chilli Powder 100 g	MTR Turmeric Powder 100 g	38
Oral Care	Closeup Everfresh Red Hot Toothpaste 43 g	Colgate Zig Zag Black Toothbrush Medium 1 pc	43
Oral Care	Colgate Cibaca Toothbrush 1 pc	Colgate Sup Fulol Head Toothbrush 1 pc	41
Oral Care	Colgate Extra Clean Medium Toothbrush 1 pc	Colgate Sup Fulol Head Toothbrush 1 pc	37
Oral Care	Colgate Sensitive Ultra Soft Toothbrush Pack 4 pcs	Colgate Zig Zag Black Toothbrush Medium 1 pc	37
Oral Care	Colgate Strong Teeth Toothpaste 150 g	Sensodyne Deep Clean Extra Soft Toothbrush 1 pc	36
Rice & Rice Products	Farmers Harvest Gold Basmati Rice 1 Kg	Heritage Basmathi Royal Rice 1 Kg	42
Rice & Rice Products	India Gate Basmathi Rice Dubar 1 Kg	India Gate Tibar Basmathi Rice 1 Kg	42
Rice & Rice Products	Farmers Harvest Rozana Basmati Rice 1 Kg	India Gate Tibar Basmathi Rice 1 Kg	39
Rice & Rice Products	India Gate Basmathi Rice Dubar 1 Kg	Natchiyar Boiled Rice Super Premium 25 Kg	39
Rice & Rice Products	Heritage Platinum Basmati Rice 5 Kg	India Gate Basmathi Rice Dubar 1 Kg	36
Staples	PPS Kabul White Channa 500g	White Channa SMALL	47
Staples	Aashirvaad Iodized Salt 1 Kg + Free Crystal Salt	PPS White Sooji (Rawa) 500g	41
Staples	Double Beans	PPS Black Channa Sundal 500g	41
Staples	Relish Herbal Jaggery Powder 400g	Samba Whole Wheat 1Kg	40
Staples	Madhu Instant Biriyani Mix	PPS Kabul White Channa 500g	39

2.6.8 CATEGORY-WISE COMBO LIFT

The Category-wise Combo Lift analysis highlights how effectively combo purchases contribute to the total sales volume across different product categories by comparing the share of line items and revenue generated from combo versus non-combo bills. Categories such as Fruits & Vegetables, Household Essentials, Feminine Care, and Instant Foods show a strong combo presence, with combo purchases contributing around 25% of total line items while non-combo purchases dominate with nearly 75%, indicating that these essential categories maintain steady sales even without combo offers but still benefit from targeted combo promotions. Meanwhile, categories like Breakfast Cereals, Rice & Rice Products, and Appalam & Papads also show a similar lift pattern, where combo-based transactions consistently account for about 25% of their sales share, demonstrating that combos effectively drive incremental revenue in

both daily-use and niche categories. Overall, the combo lift values suggest that while non-combo transactions form the bulk of category consumption, combo offers play a meaningful role in boosting additional sales across all categories, making them a strategic tool for increasing both volume and value.

Output:

```
# Groups:   Product_Category [13]
   Product_Category                  Combo_Opted line_items total_amount share_percent
   <chr>                             <chr>            <int>        <dbl>         <dbl>
 1 Appalam & Papads                  No               13436      1764275.          74.6
 2 Appalam & Papads                  Yes               4572       593146.          25.4
 3 Baby Needs                        No                7858      5588804.          74.9
 4 Baby Needs                        Yes               2633      1873306.          25.1
 5 Bread Spreads, Sauces & Mayonnaise No              13438      3829542.          74.6
 6 Bread Spreads, Sauces & Mayonnaise Yes              4581      1321819.          25.4
 7 Breakfast Cereals                 No               15232      3768445.          75.0
 8 Breakfast Cereals                 Yes               5081      1275169.          25.0
 9 Feminine CareÂ                    No               60971     22629856.          74.8
10 Feminine CareÂ                    Yes              20576      7620269.          25.2
11 Fruits & Vegetables               No               72853      9706367.          74.7
12 Fruits & Vegetables               Yes              24668      3351494.          25.3
13 Household Essentials              No               60716      9002754.          74.4
14 Household Essentials              Yes              20850      3082495.          25.6
15 Instant Foods                     No               13964      3689958.          74.4
16 Instant Foods                     Yes               4818      1265783.          25.6
17 Kitchen Essentials                No               19099      2570143           74.0
18 Kitchen Essentials                Yes               6727       911198           26.0
19 Masala & Spices                   No               14497      1326078.          74.5
20 Masala & Spices                   Yes               4973       450360.          25.5
21 Oral Care                         No               14518      2395370.          74.8
22 Oral Care                         Yes               4882       802432.          25.2
23 Rice & Rice Products              No                9772      8919733.          75.2
24 Rice & Rice Products              Yes               3229      3053596.          24.8
25 Staples                           No               19429      2220422           74.6
26 Staples                           Yes               6605       758024           25.4
```

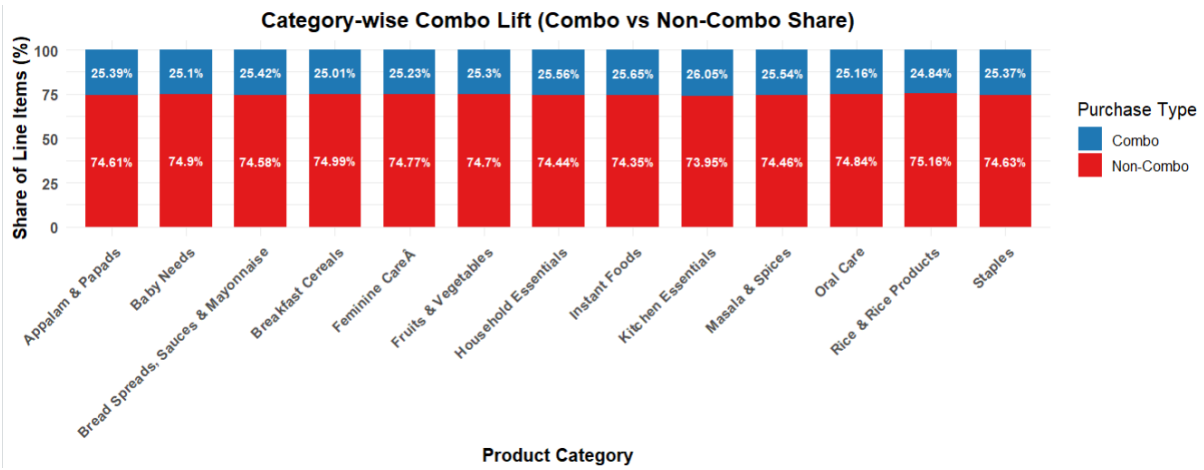

Figure 13. Category-Wise Combo lift

2.6.9 PROFITABILITY COMPARISON: COMBO VS NON-COMBO BILLS

This section compares the profitability of combo versus non-combo bills to understand how customer purchasing behaviour influences revenue outcomes. By analysing the total number of bills, the average bill value, and the overall bill amount generated under each category, this comparison helps determine whether combo offers contribute positively to sales or whether customers who opt out of combos generate higher revenue. The inclusion of both average and total bill values provides a comprehensive view of how each group impacts overall profitability and supports strategic pricing, promotional planning, and combo optimisation.

The results show that non-combo bills generate a significantly higher average bill value of ₹2,836 compared to ₹1,918 for combo bills, and they contribute a much larger total revenue of ₹7.74 crore versus ₹2.63 crore from combo transactions. Although combo bills account for a considerable number of purchases (13,740 bills), their lower average bill value suggests that customers selecting combos may be opting for budget-friendly bundled options. Overall, non-combo customers drive stronger profitability per bill, while combos primarily support customer engagement and volume rather than high-value revenue.

Coding

```
> profit_comparison <- bill_profit_summary %>%
+   group_by(Combo_Opted) %>%
+   summarise(
+     count_bills        = n(),
+     avg_bill_value     = mean(bill_value, na.rm = TRUE),
+     total_sum_bill_val = sum(bill_value, na.rm = TRUE),  # 👈 NEW COLUMN
+     .groups = "drop"
+   )
>
> profit_comparison


> ggplot(profit_comparison,
+        aes(x = Combo_Opted,
+            y = avg_bill_value,
+            fill = Combo_Opted)) +
+     geom_col() +
+     geom_text(aes(label = round(avg_bill_value, 0)),
+               vjust = -0.5, size = 3.5) +
+     labs(
+       title = "Average Bill Value by Combo Option",
+       x = "Combo Opted",
+       y = "Average Bill Value (₹)"
+     ) +
```

```
+     theme_minimal()

ggplot(profit_comparison,
+       aes(x = Combo_Opted,
+           y = total_sum_bill_val,
+           fill = Combo_Opted)) +
+     geom_col() +
+     geom_text(aes(label = round(total_sum_bill_val / 1e6, 1)),
+               vjust = -0.5, size = 3.5) +
+     labs(
+       title = "Total Revenue by Combo Option",
+       x = "Combo Opted",
+       y = "Total Bill Value (₹)"
+     ) +
+     theme_minimal()
```

Output :

```
# A tibble: 2 × 4
  Combo_Opted count_bills avg_bill_value
  <chr>           <int>         <dbl>
1 No              27292         2836.
2 Yes             13740         1918.
# ℹ 1 more variable: total_sum_bill_val <dbl>
# A tibble: 2 × 4
  Combo_Opted count_bills avg_bill_value total_sum_bill_val
  <chr>           <int>         <dbl>             <dbl>
1 No              27292         2836.         77411747.
2 Yes             13740         1918.         26359091.
```

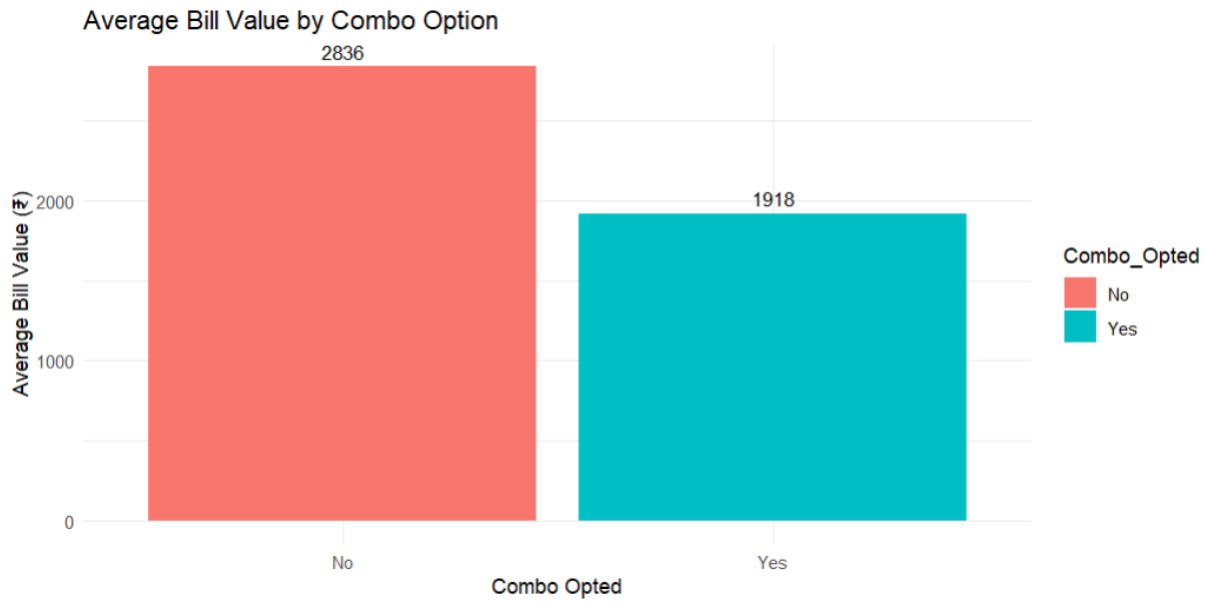

Figure 14. Average Bill Value by Combo option

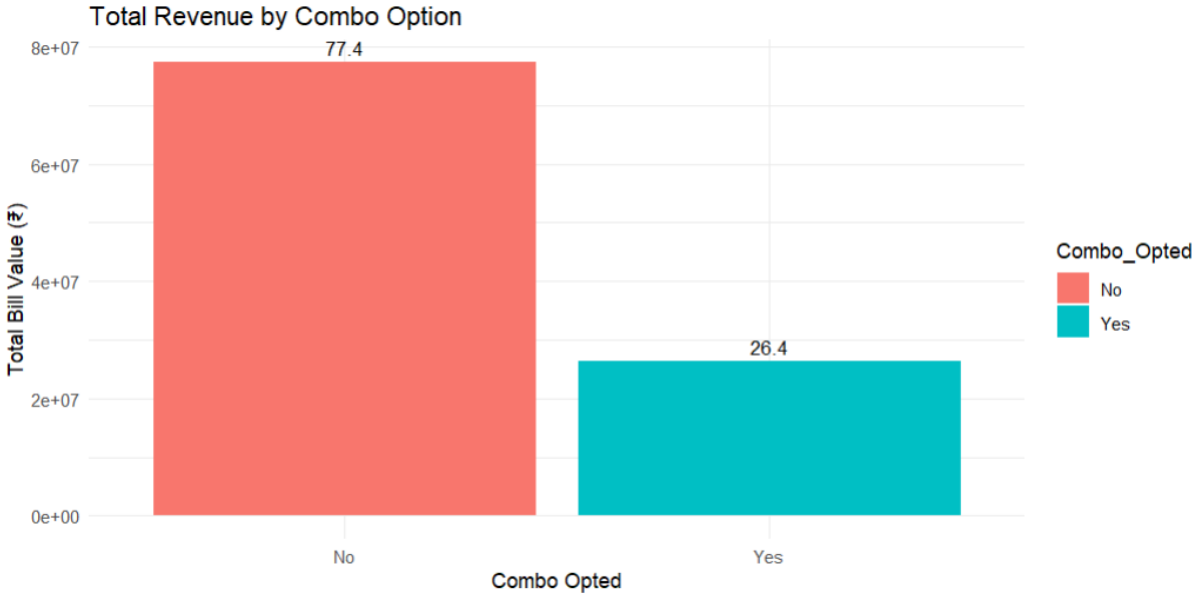

Figure 15. Total Revenue by Combo Option

2.6.10 REVNUE SHARE COMPARISON: COMBO VS NON-COMBO BILLS

This section examines the revenue share generated from combo and non-combo bills to understand how each purchase type contributes to the store's overall earnings. By comparing total bills, total revenue, and average bill value for both groups, the analysis provides a clear view of whether combo offers help increase revenue participation or whether non-combo purchases remain the primary drivers. The revenue share percentage is calculated to show the proportion of total earnings contributed by each category, offering valuable insight into how customer choices between combo and non-combo options shape overall financial performance.

Coding

```
combo_revenue_impact <- bill_main %>%
+     group_by(Combo_Opted) %>%
+     summarise(
+       total_bills   = n(),
+       total_revenue = sum(Total_Amount, na.rm = TRUE),
+       avg_bill      = mean(Total_Amount, na.rm = TRUE),
+       .groups = "drop"
+     ) %>%
+     mutate(
+       grand_total_revenue = sum(total_revenue),
+       revenue_share_pct   = round(total_revenue / grand_total_revenue * 100, 2)
+     )
>
> combo_revenue_impact
```

```
ggplot(combo_revenue_impact, aes(x = Combo_Opted, y = revenue_share_pct, fill = Combo
_Opted)) +
+     geom_col(width = 0.6) +
+     geom_text(aes(label = paste0(revenue_share_pct, "%")),
+               vjust = -0.5, fontface = "bold") +
+     labs(
+       title = "Revenue Share: Combo vs Non-Combo Bills",
+       x = "Combo Opted",
+       y = "Revenue Share (%)"
+     ) +
+     scale_fill_manual(values = c("Yes" = "#1f77b4", "No" = "#ff7f0e")) +
+     theme_minimal() +
+     theme(plot.title = element_text(face = "bold", hjust = 0.5),
+           legend.position = "none")
```

Output :

```
# A tibble: 2 × 6
  Combo_Opted total_bills total_revenue avg_bill grand_total_revenue
  <chr>           <int>        <dbl>    <dbl>            <dbl>
1 No              13552     39015285    2879.          78549300
2 Yes             13740     39534015    2877.          78549300
# ℹ 1 more variable: revenue_share_pct <dbl>
```

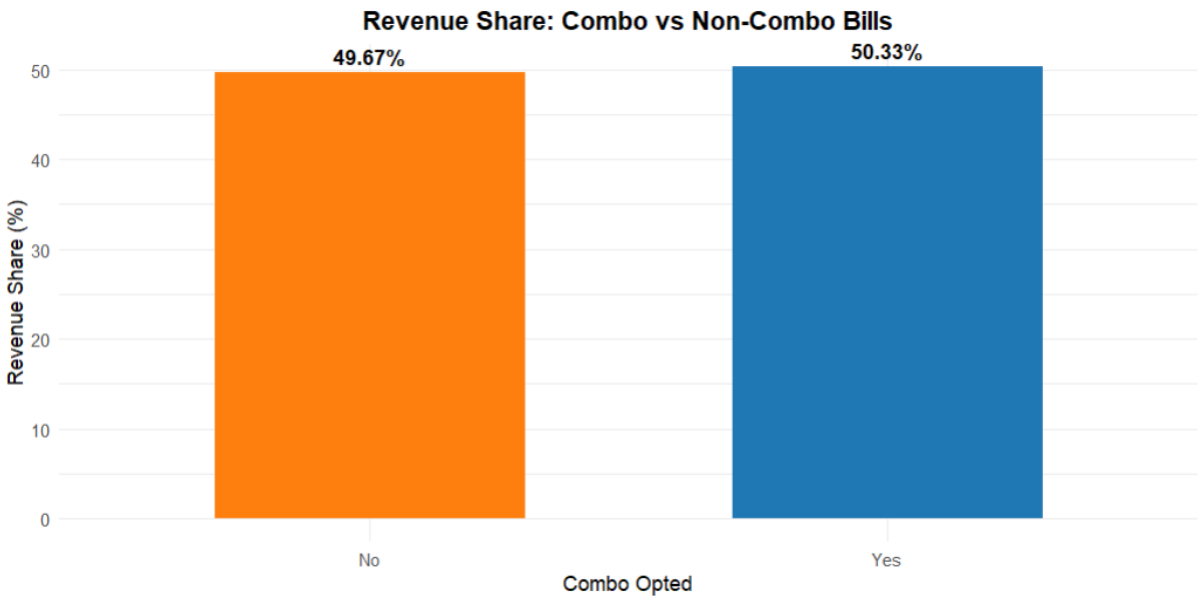

Figure 16. Revenue Share : Combo vs Non-Combo Bills

2.6.11 PRICE BAND DISTRIBUTION : COMBO VS NON-COMBO BILLS

This section analyses how products in different price bands are distributed between combo and non-combo purchases. By grouping all line items into price ranges (₹0–50, ₹51–100, ₹101–200, ₹201–500, and ₹500+), and then comparing their percentage share separately for combo and non-combo bills, the price band distribution shows whether combo offers are concentrated only in low-value items or are spread consistently across the full pricing spectrum. This helps understand customer affordability preferences and whether combo design is aligned with the store's natural price-mix.

Coding :

```
library(dplyr)
> library(ggplot2)
>
> # Ensure price_band exists
> # (You already ran this, but safe to keep once)
> bill_product <- bill_product %>%
+     mutate(
+       price_band = cut(
+         ProductRate,
+         breaks = c(0, 50, 100, 200, 500, Inf),
+         labels = c("0-50", "51-100", "101-200", "201-500", "500+"),
+         right = TRUE
+       )
+     )
>
> # Count by price band and combo flag
> price_band_combo <- bill_product %>%
+     group_by(Combo_Opted, price_band) %>%
+     summarise(n = n(), .groups = "drop") %>%
+     group_by(Combo_Opted) %>%
+     mutate(
+       percent = round(n / sum(n) * 100, 2)
+     )
>
> price_band_combo

>


> ggplot(combo_revenue_impact, aes(x = Combo_Opted, y = revenue_share_pct, fill = Comb
o_Opted)) +
+     geom_col(width = 0.6) +
+     geom_text(aes(label = paste0(revenue_share_pct, "%")),
+               vjust = -0.5, fontface = "bold") +
+     labs(
+       title = "Revenue Share: Combo vs Non-Combo Bills",
```

```
+       x = "Combo Opted",
+       y = "Revenue Share (%)"
+     ) +
+     scale_fill_manual(values = c("Yes" = "#1f77b4", "No" = "#ff7f0e")) +
+     theme_minimal() +
+     theme(plot.title = element_text(face = "bold", hjust = 0.5),
+         legend.position = "none")
>
> ggplot(price_band_combo, aes(x = price_band, y = percent, fill = Combo_Opted)) +
+     geom_col(position = "dodge", width = 0.6) +
+     geom_text(aes(label = paste0(percent, "%")),
+               position = position_dodge(width = 0.6),
+               vjust = -0.5, size = 3, fontface = "bold") +
+     labs(
+       title = "Price Band Distribution: Combo vs Non-Combo Items",
+       x = "Price Band (₹)",
+       y = "Percentage of Line Items",
+       fill = "Combo Opted"
+     ) +
+     theme_minimal() +
+     theme(plot.title = element_text(face = "bold", hjust = 0.5))
```

Output :

```
# A tibble: 10 × 4
# Groups:   Combo_Opted [2]
   Combo_Opted price_band      n percent
   <chr>       <fct>       <int>   <dbl>
 1 No          0-50       140858    42.0
 2 No          51-100      80056    23.8
 3 No          101-200     64601    19.2
 4 No          201-500     42972    12.8
 5 No          500+         7296     2.17
 6 Yes         0-50        48002    42.0
 7 Yes         51-100      27306    23.9
 8 Yes         101-200     21910    19.2
 9 Yes         201-500     14505    12.7
10 Yes         500+         2472     2.16
```

INFERENCE : The results show an almost identical price band pattern for both combo and non-combo items, indicating that combos are well integrated across the normal pricing structure. For non-combo items, ₹0–50 accounts for 42.0% (1,40,858 line items), ₹51–100 for 23.8% (80,056), ₹101–200 for 19.2% (64,601), ₹201–500 for 12.8% (42,972), and ₹500+ for 2.17% (7,296). Combo items follow the same structure, with ₹0–50 at 42.0% (48,002), ₹51–100 at 23.9% (27,306), ₹101–200 at 19.2% (21,910), ₹201–500 at 12.7% (14,505), and ₹500+ at 2.16% (2,472). This near-perfect alignment shows that combo offers are not skewed toward

only very low-priced products but reflect the natural spending pattern of customers across all key price ranges, supporting balanced and customer-friendly combo design.

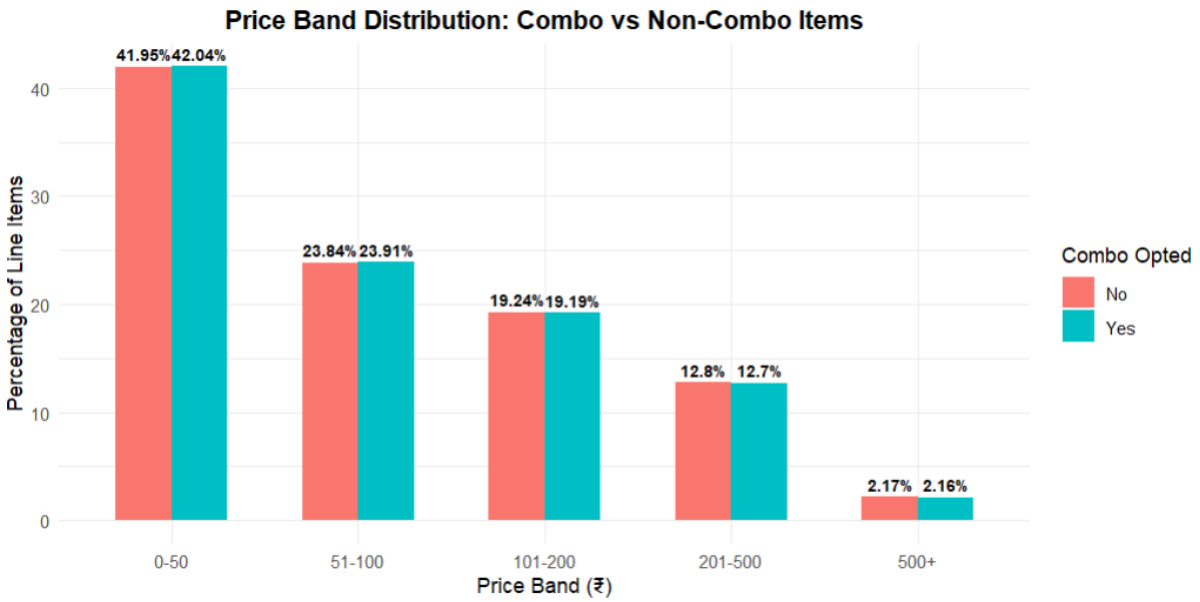

Figure 17. Price Band Distribution : Combo vs Non-Combo Bills

2.6.12 PRODUCT-LEVEL CONTRIBUTION OF COMBO PURCHASES

This section evaluates the product-level contribution of combo purchases to identify which individual items drive the highest combo-based demand. By calculating total quantity sold, quantity sold through combos, and the share of combo purchases for each product, the analysis highlights how strongly each item responds to combo-driven sales. The metrics such as combo share within a product and combo share within all combo purchases help understand whether combos influence buying behaviour consistently across products or if they mainly boost sales for specific items. This breakdown is essential for optimizing combo design, identifying high-impact products, and improving category-level merchandising strategies.

Coding :

```
product_combo_contribution <- bill_product %>%
+     group_by(Product_Name) %>%
+     summarise(
+         total_qty      = sum(Number_of_Qty, na.rm = TRUE),
+         combo_qty      = sum(ifelse(Combo_Opted == "Yes", Number_of_Qty, 0), na.rm = TR
UE),
+         noncombo_qty   = sum(ifelse(Combo_Opted == "No",  Number_of_Qty, 0), na.rm = T
RUE),
+         combo_lines    = sum(Combo_Opted == "Yes", na.rm = TRUE),
+         total_lines    = n(),
```

```
+         .groups = "drop"
+     ) %>%
+     filter(total_qty > 0) %>%
+     mutate(
+         combo_share_within_product = round(combo_qty / total_qty * 100, 2),
+         combo_share_within_all_combo = round(combo_qty / sum(combo_qty) * 100, 4) * 10
0
+     ) %>%
+     arrange(desc(combo_qty))
>
> head(product_combo_contribution, 50)


> print(product_combo_contribution,n=50)
```

Output :

Product_Name	total_qty	combo_qty	noncombo_qty	combo_lines	total_lines	combo_share_within_product	combo_share_within_all_combo
Maaretti Bruschette Roasted Garlic Bread 70 g	3252	784	2468	384	1613	24.11	34.31
Whisper Ultra Soft Air Fresh XL Pads 30 Pads	1755	519	1236	259	867	29.57	22.71
Godrej Aer Power Pocket Fresh Blossom 10 gm	1731	504	1227	251	878	29.12	22.05
Stayfree Secure Cottony 12 Pads	1775	491	1284	231	860	27.66	21.49
Del Monte Eggless Mayo 80 g	1673	488	1185	239	848	29.17	21.35
Harima Whipping Cream Powder 50 g	1688	485	1203	233	829	28.73	21.22
Aashirvaad Iodized Salt 1 Kg + Free Crystal Salt	1687	481	1206	234	834	28.51	21.05
Om Shanthi Pure Kumkum 50 gm	1755	480	1275	237	864	27.35	21
Santoor Bath Soap Pureglow 3N*125g	1700	480	1220	240	845	28.24	21
Whisper Bindazzz Nights Koala Soft XXXL + 4 Pads 1 No	1640	480	1160	234	815	29.27	21
Aachi Chilli Powder 50 g	1772	478	1294	247	876	26.98	20.92

Sago Rice Papad Punjabi Special 100 gm	1613	478	1135	239	817	29.63	20.92
Coccinia 500g	1690	477	1213	239	855	28.22	20.87
India Gate Basmathi Rice Dubar 1 Kg	1739	477	1262	238	872	27.43	20.87
PPS Sugar (1 Kg)	1699	477	1222	237	844	28.08	20.87
Green Chilli 100g	1688	476	1212	232	844	28.2	20.83
Mahali Potato 1Kg	1800	476	1324	238	906	26.44	20.83
Godrej Aer Power Pocket Lavender Bloom 10 gm	1654	475	1179	224	824	28.72	20.79
Country Pumpkin 500g	1735	474	1261	230	872	27.32	20.74
All Out Sattva Refill 45 ml	1734	473	1261	233	848	27.28	20.7
Longan Fruit 500g	1680	473	1207	229	838	28.15	20.7
Whisper Choice Wings 6 Pads	1713	473	1240	237	865	27.61	20.7
Funfoods Veg Mayonnaise Original 800 g	1759	472	1287	246	886	26.83	20.65
Sensodyne Sensitive Toothbrush Offer Pack 4 pcs	1587	472	1115	232	797	29.74	20.65
Green Apple 1Kg	1650	470	1180	231	826	28.48	20.57
Veg Greens Moringa 1 Bunch	1719	470	1249	230	866	27.34	20.57
Anjeer Fruit 1 Box	1666	469	1197	231	834	28.15	20.52
Mangaldeep Treya 3 in 1 Sambrani Sticks 18 Pcs	1620	469	1151	239	804	28.95	20.52
Whisper Thick Nights XXXL 10 Pads 1 No	1623	469	1154	226	791	28.9	20.52
Parwal 250g	1758	467	1291	235	872	26.56	20.44
Tamarind Dosa Ã Â®Â¤Ã Â¯â€¹Ã Â ®Å¡Ã Â¯Ë† Ã Â®Âªã Â¯Â • Ã Â ®Â³Ã Â®Â¿	1668	467	1201	225	823	28	20.44

Duracell Recharge AAA Ultra Batteries 2 Pcs	1689	466	1223	228	832	27.59	20.39
Horse Gram Ã Â®â€¢Ã Â¯Å Ã Â ®Â³Ã Â¯Â • Ã Â®Â³ Ã Â¯Â •	1554	466	1088	223	768	29.99	20.39
L.G Asafoetida Cake 100 g	1651	466	1185	232	820	28.23	20.39
Snacatac Cornado Cheese Burst Tin 40 g	1646	466	1180	233	828	28.31	20.39
Peacock Chandan Tika Pooja Powder 50 gm	1601	465	1136	219	790	29.04	20.35
Popular Masala Pappad 200 gm	1736	465	1271	233	871	26.79	20.35
Maharaja Orid Dhall 1Kg	1695	464	1231	227	827	27.37	20.3
Nestle Cerelac 5 Grains & Fruits Stage 5 300 g	1727	464	1263	230	855	26.87	20.3
Onion 500g	1762	464	1298	229	883	26.33	20.3
Banana Flower 1 Pc	1724	463	1261	233	868	26.86	20.26
Marvetti Oven Baked Bruschette Chips 70 g	1661	463	1198	224	839	27.87	20.26
Normalife Papaya Lemon Chia Healthy Fruit Spread 200 g	1749	463	1286	236	871	26.47	20.26
Wimco Homelites Big Match Box 1 No	1655	463	1192	241	843	27.98	20.26
Aachi Appalam 100 gm	1643	462	1181	235	840	28.12	20.22
Aashirvaad Samba Broken Rava Wheat 500g	1659	462	1197	225	834	27.85	20.22
McCain Chilli Garlic Potato Bites 200 g	1689	462	1227	228	841	27.35	20.22
Cauliflower 1 Pc	1716	461	1255	217	865	26.86	20.17
Mothers Recipe Rice Papad Ajwain 75 gm	1605	461	1144	224	820	28.72	20.17
Stayfree Secure XL 6 Pads	1634	461	1173	232	823	28.21	20.17
Elite Double Roasted Rava 1Kg (Free Wheat Fine 500g)	1591	460	1131	219	795	28.91	20.13
Harima Pani Puri Papad 100 gm	1541	460	1081	215	774	29.85	20.13

nua Zero Irritation Promise 30N (Variant)	1573	460	1113	220	780	29.24	20.13
Cycle Pure 3 in 1 Agarbathies 80 gm	1676	459	1217	220	831	27.39	20.09
Pomegranate Kabul 500g	1723	459	1264	224	851	26.64	20.09
Odonil Room Freshener Citrus Fresh 250 gm	1622	458	1164	218	806	28.24	20.04
Godrej Aer O Aer Rose Blossom 7.5 gm	1675	457	1218	222	832	27.28	20
Vari Kidney Beans	1678	457	1221	225	825	27.23	20
Watermelon Kiran 3Kg	1590	457	1133	223	794	28.74	20
Banana Karpooravalli 500g	1699	456	1243	222	862	26.84	19.95
Colgate Extra Clean Medium Toothbrush 1 pc	1777	456	1321	228	885	25.66	19.95
Good Knight Gold Flash Liquid 45 ml	1737	456	1281	229	861	26.25	19.95
Indu Rock Salt 1Kg	1633	456	1177	220	812	27.92	19.95
Mangaldeep 4 in 1 Agarbatti Ziplock 191 gm	1688	456	1232	234	836	27.01	19.95
Balamurugan Nool Thiri 108 Mini 1 No	1603	455	1148	233	801	28.38	19.91
Brinjal Ujala 250g	1620	455	1165	218	823	28.09	19.91
Baby Potato 1 Pack	1654	454	1200	224	819	27.45	19.87
Haiku Milk Bread 350 g	1583	454	1129	223	806	28.68	19.87
Veet Cold Wax Strips Sensitive Skin 8 Strips	1600	454	1146	222	807	28.38	19.87
Whisper Bindazzz Nights Koala Soft XXL + 5 Pads 1 No	1762	453	1309	232	897	25.71	19.82
Whisper Ultra Soft XL With Wings 15 Pads	1617	453	1164	225	808	28.01	19.82
Haiku Mushroom 1 Box	1740	452	1288	226	870	25.98	19.78
Sakthi Briyani Masala 50 g	1665	452	1213	229	847	27.15	19.78
Kuruvi Thinai 1Kg	1632	451	1181	230	816	27.63	19.73

Kwality Breakfast Cereals Combo Pouch 6Ãƒâ€”20 g	1748	450	1298	224	864	25.74	19.69
Sofy Anti Bacteria Extra Long+ Sanitary Pads 6 Pads	1703	450	1253	226	852	26.42	19.69
Chennai Express Orid Dhal 1Kg	1645	449	1196	228	828	27.29	19.65
24 Mantra Organic Creamy Peanut Butter 450 g	1687	448	1239	219	835	26.56	19.6
Delight Nuts California Almonds Roasted & Salted 200g	1646	448	1198	230	828	27.22	19.6
Eveready 1015 Carbon Zinc AA Battery 1 No	1651	448	1203	222	815	27.14	19.6
Fix Shadow SPF 50 Gel 40 g	1704	448	1256	221	848	26.29	19.6
Heritage Platinum Basmati Rice 5 Kg	1621	448	1173	218	817	27.64	19.6
Indian Plums 250g	1677	448	1229	221	843	26.71	19.6
Maharaja Instant Biriyani Paste	1682	448	1234	218	840	26.63	19.6
Camel Instant Sambrani 1 No	1623	447	1176	223	810	27.54	19.56
Mayil Mark Samba Rava 500g	1643	446	1197	213	803	27.15	19.52
PPS Black Urad Dhal Split 500g	1810	446	1364	224	897	24.64	19.52
Cluster Beans 250g	1672	445	1227	219	841	26.61	19.47
Cycle Rushi Sambrani Loban 50 gm	1636	445	1191	224	825	27.2	19.47
Lia Chandanam Room Freshener 160 ml	1605	445	1160	225	804	27.73	19.47
Stayfree Dry Max All Night Ultra Dry XL 18 Pads	1701	445	1256	223	849	26.16	19.47
Blueberry Box	1636	444	1192	212	793	27.14	19.43
Mint Leaves 1 Bunch	1599	444	1155	222	805	27.77	19.43
PPS Kabul White Channa 500g	1651	444	1207	220	825	26.89	19.43
Kiwi 3 pcs	1712	443	1269	228	858	25.88	19.38
Mangaldeep Sambrani Cups 12 No	1685	443	1242	217	827	26.29	19.38

Normalife Guava Chilli Chia Healthy Fruit Spread 200 g	1548	443	1105	216	772	28.62	19.38
Bombay Shaving Company Sensi Smart 3 Razor 1 Unit	1607	442	1165	218	801	27.5	19.34
Krishna 3D Fryams 100 gm	1714	442	1272	217	863	25.79	19.34
Milky Mist Gulab Jamun Tin 500 g	1656	442	1214	224	820	26.69	19.34

2.6.13 INVENTORY RISK INDICATOR FOR SLOW-MOVING PRODUCTS

This section identifies slow-moving products that pose a potential inventory risk, especially those still appearing in combo purchases despite having low total sales volume. The analysis begins by summarizing each product's total quantity sold and separating this into combo and non-combo quantities. Using the 25th percentile as a cutoff, products falling into the bottom quartile of total sales are classified as slow movers. Among these slow movers, the model highlights only those that still contribute measurable combo quantities, revealing items that might be getting artificially pushed through combos but continue to lag in natural demand. This helps identify products requiring attention in inventory planning, SKU rationalization, and promotional recalibration to avoid overstocking, excess holding costs, and inefficient shelf space utilization.

Coding :

```
> # Step 1: base summary per product
> product_speed <- bill_product %>%
+     group_by(Product_Name) %>%
+     summarise(
+         total_qty    = sum(Number_of_Qty, na.rm = TRUE),
+         combo_qty    = sum(ifelse(Combo_Opted == "Yes", Number_of_Qty, 0), na.rm = TR
UE),
+         noncombo_qty = sum(ifelse(Combo_Opted == "No",  Number_of_Qty, 0), na.rm = T
RUE),
+         .groups = "drop"
+     )
>
> # Step 2: identify slow movers (bottom 25 percent by total_qty)
> qty_threshold <- quantile(product_speed$total_qty, 0.25, na.rm = TRUE)
>
> inventory_risk_products <- product_speed %>%
+     mutate(
+         combo_share = ifelse(total_qty > 0, round(combo_qty / total_qty * 100, 2), 0),
+         slow_mover  = total_qty <= qty_threshold
+     ) %>%
```

```
+     filter(slow_mover == TRUE, combo_qty > 0) %>%   # used in combos but still slow
+     arrange(total_qty)
>
> head(inventory_risk_products, 50)
```

Output :

```
# A tibble: 50 × 6
   Product_Name                        total_qty combo_qty noncombo_qty combo_share slow_mov
er
   <chr>                               <dbl>     <dbl>       <dbl>       <dbl> <lgl>
 1 Godrej Aer Spray Violet Valley Bloo…     1421       377        1044        26.5 TRUE
 2 Guava Imported 1Kg                       1460       370        1090        25.3 TRUE
 3 Sweet Potato 500g                        1465       352        1113        24.0 TRUE
 4 Kelloggs Chocos 110 g                    1480       401        1079        27.1 TRUE
 5 Colgate Zig Zag Toothbrush Pack of 6     1483       364        1119        24.5 TRUE
 6 Kelloggs Oats Pouch 200 g                1486       419        1067        28.2 TRUE
 7 Godrej Aer Twist Gel Bright Tangy D…     1488       371        1117        24.9 TRUE
 8 Meena Cotton Wicks 1 Pack                1488       394        1094        26.5 TRUE
 9 Chow Chow 500g                           1492       370        1122        24.8 TRUE
10 Godrej Aer Pocket Bathroom Freshene…     1492       406        1086        27.2 TRUE
# ℹ 40 more rows
# ℹ Use `print(n = ...)` to see more rows
>
> print(inventory_risk_products,n = 50)
# A tibble: 138 × 6
   Product_Name                        total_qty combo_qty noncombo_qty combo_share slow_mov
er
   <chr>                               <dbl>     <dbl>       <dbl>       <dbl> <lgl>
 1 "Godrej Aer Spray Violet Valley Blo…     1421       377        1044        26.5 TRUE
 2 "Guava Imported 1Kg"                     1460       370        1090        25.3 TRUE
 3 "Sweet Potato 500g"                      1465       352        1113        24.0 TRUE
 4 "Kelloggs Chocos 110 g"                  1480       401        1079        27.1 TRUE
 5 "Colgate Zig Zag Toothbrush Pack of…     1483       364        1119        24.5 TRUE
 6 "Kelloggs Oats Pouch 200 g"              1486       419        1067        28.2 TRUE
 7 "Godrej Aer Twist Gel Bright Tangy …     1488       371        1117        24.9 TRUE
 8 "Meena Cotton Wicks 1 Pack"              1488       394        1094        26.5 TRUE
 9 "Chow Chow 500g"                         1492       370        1122        24.8 TRUE
10 "Godrej Aer Pocket Bathroom Freshen…     1492       406        1086        27.2 TRUE
11 "Whisper Koala Soft XXXL Sanitary P…     1499       384        1115        25.6 TRUE
12 "Colgate Sensitive Ultra Soft Tooth…     1503       356        1147        23.7 TRUE
13 "Odonil Blocks Jasmine Mist 50 gm"       1503       401        1102        26.7 TRUE
14 "Greens Fenugreek 1 Bunch"               1504       387        1117        25.7 TRUE
15 "Bitter Gourd Hybrid 500g"               1506       378        1128        25.1 TRUE
16 "Bella Regular Soft Wings 8 Pads 1 …     1508       407        1101        27.0 TRUE
17 "Virali Manjal 250g"                     1508       379        1129        25.1 TRUE
18 "Bella Perfecta Ultra Maxi Blue XL …     1510       404        1106        26.8 TRUE
19 "Diamond Sugar (Kalkandu) 250g"          1512       408        1104        27.0 TRUE
20 "Disano Oats 1 Kg"                       1513       341        1172        22.5 TRUE
21 "Good Knight Gold Flash Twin Pack R…     1514       410        1104        27.1 TRUE
22 "Baby Corn 1 Pack"                       1518       373        1145        24.6 TRUE
```

23 "Aachi Kulambu Chilly Masala 100 g" 1520 396 1124 26.0 TRUE
24 "Banana Green 500g" 1520 380 1140 25 TRUE
25 "Good Knight Gold Flash Refills 4Ã—… 1520 387 1133 25.5 TRUE
26 "Sakthi Chicken Masala 50 g" 1520 408 1112 26.8 TRUE
27 "Gopuram Thazhampoo Kumkum 40 gm" 1523 380 1143 25.0 TRUE
28 "KSC Chilli Long 200 g" 1523 343 1180 22.5 TRUE
29 "Nestle Cerelac Wheat Apple Cherry … 1524 372 1152 24.4 TRUE
30 "Dabur Hommade Tomato Puree 200 g" 1525 391 1134 25.6 TRUE
31 "Bindu Gold Applam Pouch 100 gm" 1528 360 1168 23.6 TRUE
32 "Puro Healthy Salt 1Kg" 1528 414 1114 27.1 TRUE
33 "Odonil Air Freshener Night Queen &… 1529 379 1150 24.8 TRUE
34 "Garlic 500g" 1530 415 1115 27.1 TRUE
35 "Kelloggs Chocos Fills 180 g" 1531 412 1119 26.9 TRUE
36 "Pear Green Howell 500g" 1531 386 1145 25.2 TRUE
37 "Tiger Pachai Karpooram 2 gm" 1531 396 1135 25.9 TRUE
38 "Whisper Ultra Hygiene 3 Smart Laye… 1534 421 1113 27.4 TRUE
39 "Samba Whole Wheat 1Kg" 1535 381 1154 24.8 TRUE
40 "Aachi Garam Masala 50 g" 1537 397 1140 25.8 TRUE
41 "Kai Beauty Beauty Box 1 No" 1540 395 1145 25.6 TRUE
42 "Harima Pani Puri Papad 100 gm" 1541 460 1081 29.8 TRUE
43 "Quaker Oats 400 g" 1541 391 1150 25.4 TRUE
44 "Snake Gourd 500g" 1541 394 1147 25.6 TRUE
45 "Stayfree Advanced All Night Ultra … 1541 437 1104 28.4 TRUE
46 "Kambu Kurunai Broken 500g (à®•à®®à… 1542 406 1136 26.3 TRUE
47 "Sago Cumin Papad 100 gm" 1544 395 1149 25.6 TRUE
48 "Cycle Sandalum Cones 10 No" 1545 379 1166 24.5 TRUE
49 "Peas Dhall à®µà®Ÿà¯^ à®ªà®°à¯\u008… 1545 405 1140 26.2 TRUE
50 "Godrej Aer Power Pocket Floral Del… 1546 349 1197 22.6 TRUE
ℹ 88 more rows
ℹ Use `print(n = .

2.6.14 COMBO DISCOUNT DEPENDENCY INDEX

The Combo Discount Dependency Index measures how strongly each product's sales depend on combo-based purchases, helping identify items that rely heavily on discounts or bundled offers to achieve movement. By calculating the percentage of each product's total quantity that comes specifically from combo transactions, the analysis reveals products with high combo dependency and flags them as discount-driven items. Products are then sorted by this dependency percentage, and a threshold (such as 60%) can be applied to classify those that are highly reliant on combo discounts. This metric is crucial for understanding which SKUs may struggle to sell at full price, guiding pricing strategy, margin protection, and decisions on whether certain products should continue in combo promotions, be repositioned, or require deeper marketing support.

Coding :

```
combo_dependency <- product_speed %>%   # reuse from previous block
+     mutate(
+       total_qty_all = combo_qty + noncombo_qty,
+       combo_dependency_pct = ifelse(
+         total_qty_all > 0,
+         round(combo_qty / total_qty_all * 100, 2),
+         0
+       ),
+       discount_dependent = combo_dependency_pct > 60   # threshold can be changed
+     ) %>%
+     arrange(desc(combo_dependency_pct))
>
> head(combo_dependency, 50)
```

```
combo_dependency <- product_speed %>%   # reuse from previous block
+     mutate(
+       total_qty_all = combo_qty + noncombo_qty,
+       combo_dependency_pct = ifelse(
+         total_qty_all > 0,
+         round(combo_qty / total_qty_all * 100, 2),
+         0
+       ),
+       discount_dependent = combo_dependency_pct > 60   # threshold can be changed
+     ) %>%
+     arrange(desc(combo_dependency_pct))
>
> head(combo_dependency, 50)
```

Output :

```
# A tibble: 50 × 7
   Product_Name            total_qty combo_qty noncombo_qty total_qty_all combo_dependenc
y_pct
   <chr>                     <dbl>     <dbl>       <dbl>        <dbl>             <dbl>
 1 "Horse Gram à®•à¯Šà®³à¯…      1554       466        1088         1554              30.0
 2 "Harima Pani Puri Papad…      1541       460        1081         1541              29.8
 3 "Sensodyne Sensitive To…      1587       472        1115         1587              29.7
 4 "Sago Rice Papad Punjab…      1613       478        1135         1613              29.6
 5 "Whisper Ultra Soft Air…      1755       519        1236         1755              29.6
 6 "Whisper Bindazzz Night…      1640       480        1160         1640              29.3
 7 "nua Zero Irritation Pr…      1573       460        1113         1573              29.2
 8 "Del Monte Eggless Mayo…      1673       488        1185         1673              29.2
 9 "Godrej Aer Power Pocke…      1731       504        1227         1731              29.1
10 "Peacock Chandan Tika P…      1601       465        1136         1601              29.0
# ℹ 40 more rows
# ℹ 1 more variable: discount_dependent <lgl>
# ℹ Use `print(n = ...)` to see more rows
>
```

```
> print(combo_dependency,n = 50)
# A tibble: 551 × 7
   Product_Name            total_qty combo_qty noncombo_qty total_qty_all combo_dependenc
y_pct
   <chr>                     <dbl>     <dbl>       <dbl>        <dbl>             <dbl>
 1 "Horse Gram à®•à¯Šà®³à¯…      1554       466        1088         1554              30.0
 2 "Harima Pani Puri Papad…      1541       460        1081         1541              29.8
 3 "Sensodyne Sensitive To…      1587       472        1115         1587              29.7
 4 "Sago Rice Papad Punjab…      1613       478        1135         1613              29.6
 5 "Whisper Ultra Soft Air…      1755       519        1236         1755              29.6
 6 "Whisper Bindazzz Night…      1640       480        1160         1640              29.3
 7 "nua Zero Irritation Pr…      1573       460        1113         1573              29.2
 8 "Del Monte Eggless Mayo…      1673       488        1185         1673              29.2
 9 "Godrej Aer Power Pocke…      1731       504        1227         1731              29.1
10 "Peacock Chandan Tika P…      1601       465        1136         1601              29.0
11 "Mangaldeep Treya 3 in …      1620       469        1151         1620              29.0
12 "Elite Double Roasted R…      1591       460        1131         1591              28.9
13 "Whisper Thick Nights X…      1623       469        1154         1623              28.9
14 "Watermelon Kiran 3Kg"        1590       457        1133         1590              28.7
15 "Harima Whipping Cream …      1688       485        1203         1688              28.7
16 "Godrej Aer Power Pocke…      1654       475        1179         1654              28.7
17 "Mothers Recipe Rice Pa…      1605       461        1144         1605              28.7
18 "Haiku Milk Bread 350 g"      1583       454        1129         1583              28.7
19 "Normalife Guava Chilli…      1548       443        1105         1548              28.6
20 "Aashirvaad Iodized Sal…      1687       481        1206         1687              28.5
21 "Green Apple 1Kg"             1650       470        1180         1650              28.5
22 "Balamurugan Nool Thiri…      1603       455        1148         1603              28.4
23 "Veet Cold Wax Strips S…      1600       454        1146         1600              28.4
24 "Stayfree Advanced All …      1541       437        1104         1541              28.4
25 "Snacatac Cornado Chees…      1646       466        1180         1646              28.3
26 "Odonil Room Freshener …      1622       458        1164         1622              28.2
27 "Santoor Bath Soap Pure…      1700       480        1220         1700              28.2
28 "L.G Asafoetida Cake 10…      1651       466        1185         1651              28.2
29 "Coccinia 500g"               1690       477        1213         1690              28.2
30 "Stayfree Secure XL 6 P…      1634       461        1173         1634              28.2
31 "Green Chilli 100g"           1688       476        1212         1688              28.2
32 "Kelloggs Oats Pouch 20…      1486       419        1067         1486              28.2
33 "Anjeer Fruit 1 Box"          1666       469        1197         1666              28.2
34 "Longan Fruit 500g"           1680       473        1207         1680              28.2
35 "Aachi Appalam 100 gm"        1643       462        1181         1643              28.1
36 "Brinjal Ujala 250g"          1620       455        1165         1620              28.1
37 "PPS Sugar (1 Kg)"            1699       477        1222         1699              28.1
38 "Whisper Ultra Soft XL …      1617       453        1164         1617              28.0
39 "Tamarind Dosa à®¤à¯‹à®…      1668       467        1201         1668                28
40 "Saffron Kunguma Poo 1g…      1569       439        1130         1569              28.0
41 "Wimco Homelites Big Ma…      1655       463        1192         1655              28.0
42 "Indu Rock Salt 1Kg"          1633       456        1177         1633              27.9
43 "Marvetti Oven Baked Br…      1661       463        1198         1661              27.9
44 "Aashirvaad Samba Broke…      1659       462        1197         1659              27.8
45 "Darling Idly Chilly Po…      1576       438        1138         1576              27.8
```

```
46 "Fixderma Nigrifix Crea…     1587     441     1146     1587          27.8
47 "Mint Leaves 1 Bunch"        1599     444     1155     1599          27.8
48 "Thulir Sanitary Napkin…     1586     440     1146     1586          27.7
49 "Lia Chandanam Room Fre…     1605     445     1160     1605          27.7
50 "Dragon Fruit 500g"          1556     431     1125     1556          27.7
```

2.6.15 SEASONAL PATTERNS IN COMBO EFFECTIVENESS

This section examines how combo effectiveness changes across months by tracking the monthly combo conversion rate, calculated as the percentage of bills that include a combo purchase. By grouping bill data by year and month, the analysis highlights seasonal variations and recurring behavioural patterns in customer adoption of combo offers. The results show a relatively stable conversion rate, mostly fluctuating between 48 percent and 53 percent, with certain months such as July 2023, February 2024, March 2024, April 2024, November 2024, April 2025, and September 2025 showing slightly higher conversion peaks. These consistent seasonal shifts indicate that customer responsiveness to combos increases during specific periods, possibly influenced by festivals, salary cycles, product availability, or promotional campaigns.

Coding :

```
> bill_main_monthly <- bill_main %>%
+     mutate(
+         BillDate = as.Date(BillDate),
+         YearMonth = format(BillDate, "%Y-%m")
+     ) %>%
+     group_by(YearMonth) %>%
+     summarise(
+         total_bills  = n(),
+         combo_bills  = sum(Combo_Opted == "Yes", na.rm = TRUE),
+         noncombo_bills = sum(Combo_Opted == "No", na.rm = TRUE),
+         combo_conversion_rate = round(combo_bills / total_bills * 100, 2),
+         .groups = "drop"
+     )
>
> bill_main_monthly

ggplot(bill_main_monthly, aes(x = YearMonth, y = combo_conversion_rate, group = 1)) +
+     geom_line(size = 1, color = "#1f77b4") +
+     geom_point(size = 2, color = "#ff7f0e") +
+     labs(
+         title = "Monthly Combo Conversion Rate",
+         x = "Month",
+         y = "Combo Conversion Rate (%)"
+     ) +
```

```
+     theme_minimal() +
+     theme(
+         axis.text.x = element_text(angle = 45, hjust = 1),
+         plot.title  = element_text(face = "bold", hjust = 0.5)
+     )
```

Output:

```
# A tibble: 30 × 5
   YearMonth total_bills combo_bills noncombo_bills combo_conversion_rate
   <chr>         <int>       <int>         <int>                 <dbl>
 1 2023-04         904         456           448                  50.4
 2 2023-05         922         425           497                  46.1
 3 2023-06         874         431           443                  49.3
 4 2023-07         954         491           463                  51.5
 5 2023-08         910         458           452                  50.3
 6 2023-09         921         454           467                  49.3
 7 2023-10         898         451           447                  50.2
 8 2023-11         889         448           441                  50.4
 9 2023-12         919         457           462                  49.7
10 2024-01         934         449           485                  48.1
# ℹ 20 more rows
# ℹ Use `print(n = ...)` to see more rows
>
> print(bill_main_monthly,n=30)
# A tibble: 30 × 5
   YearMonth total_bills combo_bills noncombo_bills combo_conversion_rate
   <chr>         <int>       <int>         <int>                 <dbl>
 1 2023-04         904         456           448                  50.4
 2 2023-05         922         425           497                  46.1
 3 2023-06         874         431           443                  49.3
 4 2023-07         954         491           463                  51.5
 5 2023-08         910         458           452                  50.3
 6 2023-09         921         454           467                  49.3
 7 2023-10         898         451           447                  50.2
 8 2023-11         889         448           441                  50.4
 9 2023-12         919         457           462                  49.7
10 2024-01         934         449           485                  48.1
11 2024-02         854         441           413                  51.6
12 2024-03         959         496           463                  51.7
13 2024-04         931         494           437                  53.1
14 2024-05         890         436           454                  49.0
15 2024-06         915         476           439                  52.0
16 2024-07         924         466           458                  50.4
17 2024-08         915         469           446                  51.3
18 2024-09         859         429           430                  49.9
19 2024-10         948         456           492                  48.1
20 2024-11         894         463           431                  51.8
21 2024-12         918         468           450                  51.0
22 2025-01         935         458           477                  49.0
```

23	2025-02	804	407	397	50.6
24	2025-03	947	484	463	51.1
25	2025-04	896	472	424	52.7
26	2025-05	926	461	465	49.8
27	2025-06	908	463	445	51.0
28	2025-07	912	437	475	47.9
29	2025-08	960	480	480	50
30	2025-09	872	464	408	53.2

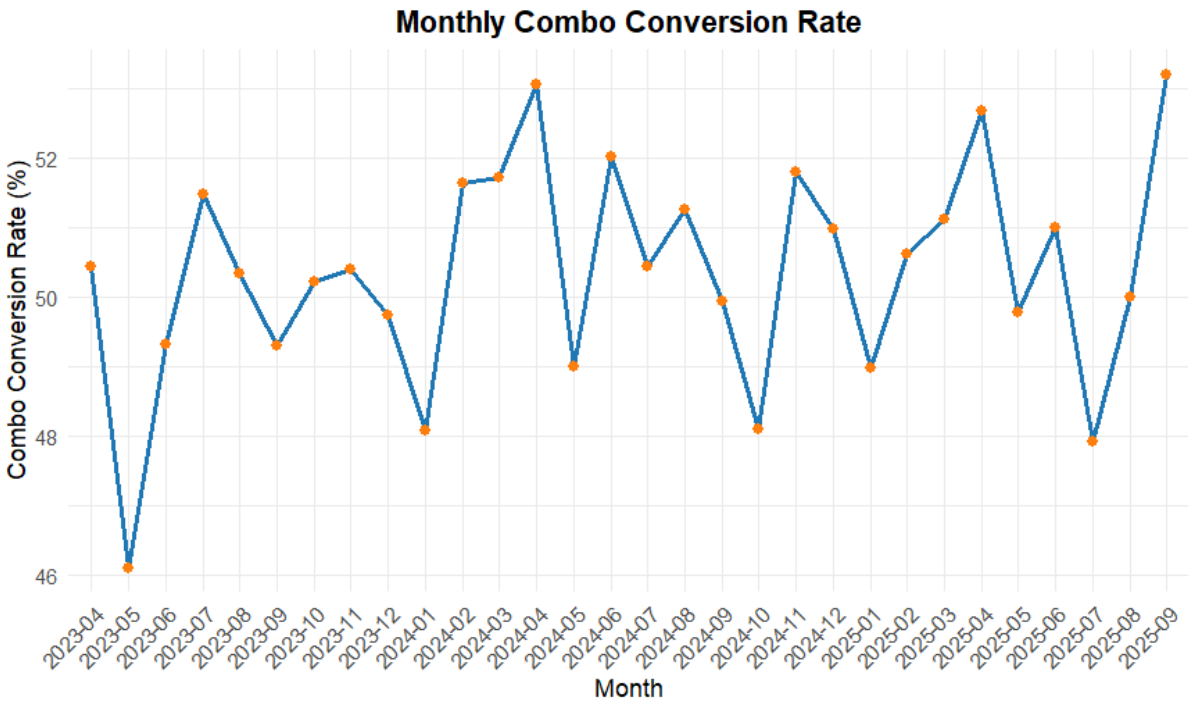

Figure 18. Monthly Combo Conversion Rate

2.6.16 REPEAT PURCHASE INFLUENCE OF COMBO ITEMS

This section analyzes whether combo items encourage repeat purchases by examining product-level sales patterns across combo and non-combo transactions. By identifying whether a product has been purchased in combo bills, non-combo bills, or both, the analysis classifies each item into behavioural patterns that reflect customer buying journeys. The results show that 551 products fall under the "Combo_then_Noncombo_or_Mixed" pattern, meaning customers first purchased these items through combos and later continued buying them even without combo offers. This indicates that combos are functioning as an effective trial mechanism, pulling customers toward new or unfamiliar products and subsequently converting them into regular, repeat purchases. Overall, the pattern demonstrates that combo promotions

not only increase immediate sales but also strengthen long-term product adoption and customer loyalty.

Coding :

```
> product_repeat_pattern <- bill_product %>%
+   group_by(Product_Name) %>%
+   summarise(
+     has_combo    = any(Combo_Opted == "Yes"),
+     has_noncombo = any(Combo_Opted == "No"),
+     first_date_combo = ifelse(has_combo,
+                               as.character(min(BillDate[Combo_Opted == "Yes"], na.rm = TRUE))
,
+                               NA),
+     first_date_noncombo = ifelse(has_noncombo,
+                                  as.character(min(BillDate[Combo_Opted == "No"], na.rm = TRUE)
),
+                                  NA),
+     .groups = "drop"
+   ) %>%
+   mutate(
+     pattern = case_when(
+       has_combo & has_noncombo ~ "Combo_then_Noncombo_or_Mixed",
+       has_combo & !has_noncombo ~ "Only_Combo",
+       !has_combo & has_noncombo ~ "Only_Noncombo",
+       TRUE ~ "No_Sales"
+     )
+   )
>
> table(product_repeat_pattern$pattern)

Combo_then_Noncombo_or_Mixed
                         551
> head(product_repeat_pattern, 50)
print(combo_trial_products, width = Inf)
# A tibble: 551 × 6
   Product_Name                                  has_combo has_noncombo first_date_combo
   <chr>                                         <lgl>     <lgl>        <chr>
 1 24 Mantra Organic Creamy Peanut Butter 450 g  TRUE      TRUE         2023-04-02
 2 Aachi Appalam 100 gm                          TRUE      TRUE         2023-04-03
 3 Aachi Chilli Powder 50 g                      TRUE      TRUE         2023-04-01
 4 Aachi Garam Masala 50 g                       TRUE      TRUE         2023-04-01
 5 Aachi Hyderabadi Biriyani 375 g               TRUE      TRUE         2023-04-01
 6 Aachi Hyderabadi Biryani Masala Paste 177 g   TRUE      TRUE         2023-05-02
 7 Aachi Kulambu Chilly Masala 100 g             TRUE      TRUE         2023-04-05
 8 Aachi Turmeric Powder 50 g                    TRUE      TRUE         2023-04-10
 9 Aashirvaad Iodized Salt 1 Kg + Free Crystal Salt TRUE   TRUE         2023-04-02
10 Aashirvaad Samba Broken Rava Wheat 500g       TRUE      TRUE         2023-04-06
   first_date_noncombo pattern
```

	<chr>	<chr>
1	2023-04-03	Combo_then_Noncombo_or_Mixed
2	2023-04-02	Combo_then_Noncombo_or_Mixed
3	2023-04-03	Combo_then_Noncombo_or_Mixed
4	2023-04-03	Combo_then_Noncombo_or_Mixed
5	2023-04-03	Combo_then_Noncombo_or_Mixed
6	2023-04-02	Combo_then_Noncombo_or_Mixed
7	2023-04-02	Combo_then_Noncombo_or_Mixed
8	2023-04-01	Combo_then_Noncombo_or_Mixed
9	2023-04-01	Combo_then_Noncombo_or_Mixed
10	2023-04-02	Combo_then_Noncombo_or_Mixed

2.6.17 Forecasting (ARIMA & Prophet)

The forecasting section evaluates the future trajectory of monthly combo revenue by applying two powerful time-series models: ARIMA and Facebook Prophet. First, monthly revenue generated exclusively from combo bills is aggregated over a 30-month period, showing stable but fluctuating performance between ₹11.9 lakh and ₹14.8 lakh per month. This historical trend is converted into a structured time series and fed into the ARIMA model, which identifies the overall mean revenue pattern and generates a six-month projection. The Prophet model is then applied to the same data to capture any underlying trend and seasonal components. Both models collectively provide a forward-looking view of expected revenue stability, potential growth phases, and demand volatility, helping the business plan procurement, pricing, and promotional campaigns more confidently.

The ARIMA forecasts indicate a continuation of the current stable revenue pattern, with projected values remaining close to the long-term mean of approximately ₹13.1 lakh. Prophet's output reinforces this stability while also providing additional decomposition insights such as monthly trend movements and seasonal influence. Together, these forecasts highlight that combo revenue is consistently dependable without sharp upward or downward spikes, making combos a predictable and strategic revenue stream. The integration of k-means–based bill segmentation further strengthens this analysis by identifying customer groups with varying spending levels, item counts, and combo adoption patterns, enabling targeted marketing interventions that align with the predicted revenue behaviour.

2.6.17.1 Prepare monthly combo revenue

Coding :

```
> library(dplyr)
> library(lubridate)
>
> # Make sure BillDate is Date
> bill_main <- bill_main %>%
+     mutate(BillDate = as.Date(BillDate))
>
> # Monthly combo revenue (example: total revenue from combo bills)
> combo_monthly <- bill_main %>%
+     mutate(YearMonth = floor_date(BillDate, unit = "month")) %>%
+     group_by(YearMonth) %>%
+     summarise(
+         combo_revenue = sum(ifelse(Combo_Opted == "Yes", Total_Amount, 0), na.rm = TR
UE),
+         .groups = "drop"
+     )
> combo_monthly
```

Output :

```
# A tibble: 30 × 2
   YearMonth  combo_revenue
   <date>             <dbl>
 1 2023-04-01       1321589
 2 2023-05-01       1191514
 3 2023-06-01       1281809
 4 2023-07-01       1429948
 5 2023-08-01       1330254
 6 2023-09-01       1301608
 7 2023-10-01       1272684
 8 2023-11-01       1273045
 9 2023-12-01       1293686
10 2024-01-01       1301426
# ℹ 20 more rows
# ℹ Use `print(n = ...)` to see more rows
>
> print(combo_monthly,n=30)
# A tibble: 30 × 2
   YearMonth  combo_revenue
   <date>             <dbl>
 1 2023-04-01       1321589
 2 2023-05-01       1191514
 3 2023-06-01       1281809
 4 2023-07-01       1429948
 5 2023-08-01       1330254
 6 2023-09-01       1301608
 7 2023-10-01       1272684
 8 2023-11-01       1273045
```

```
 9 2023-12-01      1293686
10 2024-01-01      1301426
11 2024-02-01      1271723
12 2024-03-01      1482326
13 2024-04-01      1430111
14 2024-05-01      1268727
15 2024-06-01      1352069
16 2024-07-01      1386057
17 2024-08-01      1307220
18 2024-09-01      1237244
19 2024-10-01      1358924
20 2024-11-01      1324378
21 2024-12-01      1345767
22 2025-01-01      1305346
23 2025-02-01      1154046
24 2025-03-01      1398466
25 2025-04-01      1295029
26 2025-05-01      1359678
27 2025-06-01      1334987
28 2025-07-01      1242075
29 2025-08-01      1361281
30 2025-09-01      1320998
```

2.6.17.2 ARIMA FORECAST (NEXT 6 MONTHS)

The ARIMA modelling process converts the monthly combo revenue into a structured time-series format and automatically fits the best-suited model using auto.arima(). The selected model, ARIMA(0,0,0) with a non-zero mean, indicates that the combo revenue follows a stable pattern without strong trends or seasonal components, and its behaviour is best represented by a constant average level rather than complex fluctuations. The model estimates an average monthly combo revenue of ₹13,17,800, with relatively low forecasting errors (MAE ≈ ₹51,322 and MAPE ≈ 3.91%). Using this fitted model, a six-month forecast is generated, producing a smooth projection that extends the existing stable pattern into the future. This helps the business anticipate revenue consistency, plan inventory, and design combo campaigns more confidently based on predictable monthly performance.

Coding:

```
library(forecast)
Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo >
```

```
> # Convert to time series (monthly)
> combo_ts <- ts(combo_monthly$combo_revenue,
+             start = c(year(min(combo_monthly$YearMonth)),
+                       month(min(combo_monthly$YearMonth))),
+             frequency = 12)
>
> # Fit ARIMA automatically
> fit_arima <- auto.arima(combo_ts)
> summary(fit_arima)
>
> # Forecast next 6 months
> fc_arima <- forecast(fit_arima, h = 6)
> plot(fc_arima,
+     main = "ARIMA Forecast of Monthly Combo Revenue",
+     xlab = "Year", ylab = "Combo Revenue")
```

Output :

```
Series: combo_ts
ARIMA(0,0,0) with non-zero mean

Coefficients:
          mean
     1317800.50
s.e.    12382.13

sigma^2 = 4.758e+09:  log likelihood = -376.31
AIC=756.61   AICc=757.06   BIC=759.42

Training set error measures:
                    ME     RMSE     MAE       MPE     MAPE      MASE       ACF1
Training set 2.244487e-07 67819.77 51321.7 -0.2664748 3.909783 0.7066363 -0.06178471
```

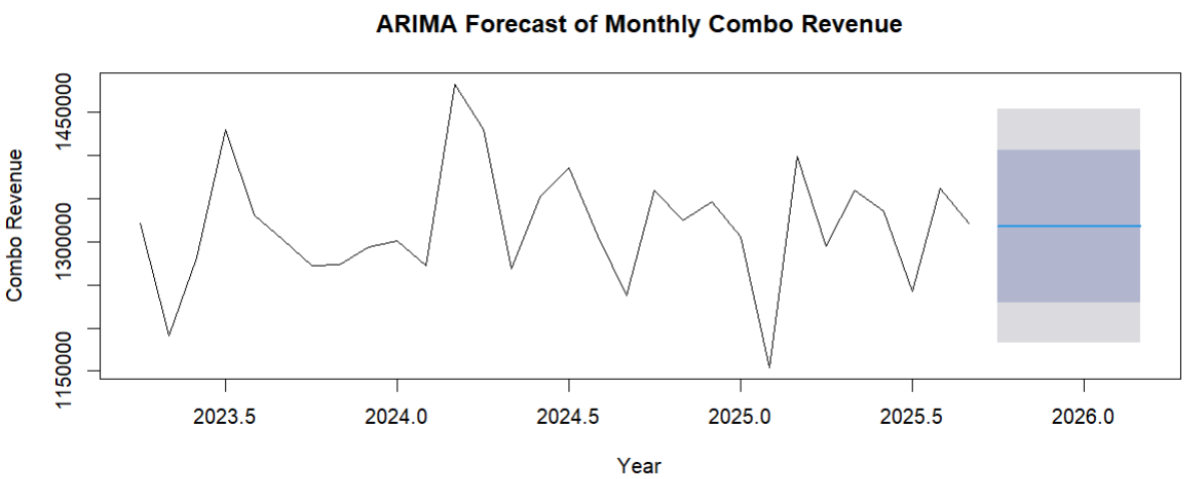

Figure 19. ARIMA Forecast of Monthly Combo Revenue

2.6.17.3 PROPHET FORECAST

The Prophet forecast provides a deeper decomposition of combo revenue into long-term trend and seasonal patterns, offering a clearer understanding of how monthly performance evolves over time. The trend component shows a steady upward movement, indicating that combo revenue is gradually increasing year after year. Meanwhile, the yearly seasonality graph highlights recurring fluctuations across months, with notable peaks and dips that repeat annually, reflecting customer behaviour influenced by festivals, climate changes, product availability, or monthly shopping cycles. The presence of strong seasonal waves combined with a consistently rising trend suggests that combo revenue is not only stable but also growing, with predictable high-demand periods that can be strategically leveraged through targeted promotions, inventory planning, and combo design.

```
# Prophet expects columns: ds (date), y (value)
> prophet_df <- combo_monthly %>%
+     rename(ds = YearMonth, y = combo_revenue)
>
> m_prophet <- prophet(prophet_df)
Disabling weekly seasonality. Run prophet with weekly.seasonality=TRUE to override this.
Disabling daily seasonality. Run prophet with daily.seasonality=TRUE to override this.
n.changepoints greater than number of observations. Using 23> future <- make_future_datafr
ame(m_prophet, periods = 6, freq = "month")
> forecast_prophet <- predict(m_prophet, future)
>
> plot(m_prophet, forecast_prophet)
> prophet_plot_components(m_prophet, forecast_prophet)
```

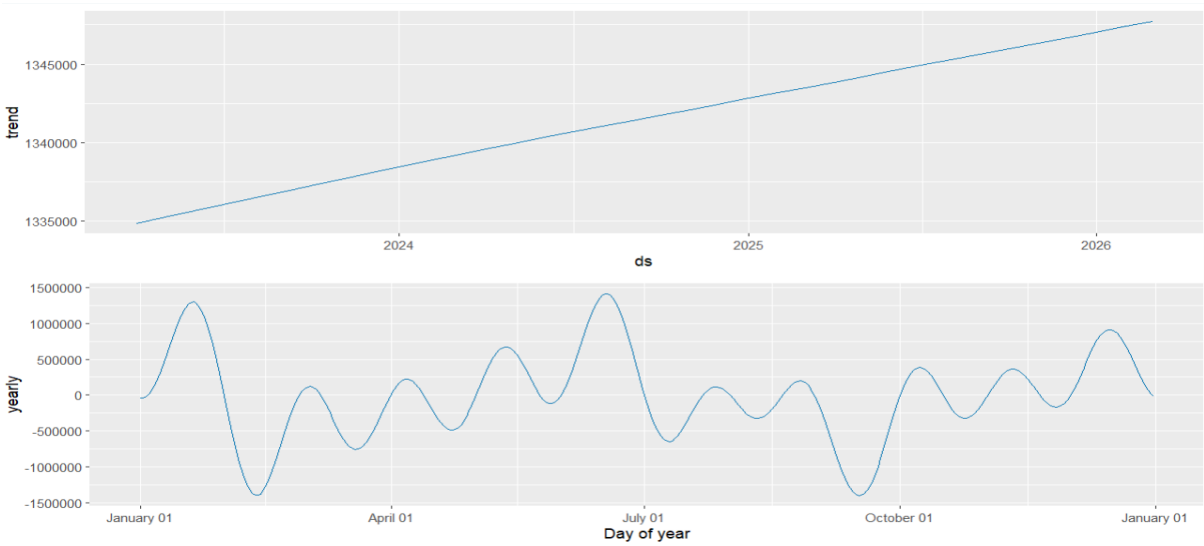

Figure 20. Prophet Forecast of Monthly Combo Revenue

2.6.17.4 CUSTOMER / BILL SEGMENTATION WITH K-MEANS

The k-means segmentation groups customer bills into meaningful clusters based on three key behavioural features: number of products purchased, total bill value, and whether the bill included a combo. After scaling the variables for comparability, the algorithm identifies three distinct customer segments, each showing unique purchasing characteristics. The first cluster represents high-value shoppers, averaging 23.7 items per bill with an average spend of ₹4,326, indicating bulk or premium purchases with moderate combo usage. The second and third clusters represent moderate and light shoppers with around 12 items per bill and average spends of ₹2,002 and ₹1,977 respectively, showing lower combo adoption and more routine purchasing behaviour. This segmentation helps the business tailor promotional strategies, combo designs, and pricing interventions to different customer groups, enabling targeted marketing and improved customer engagement.

```
> library(dplyr)
>
> # Build bill-level feature table
> bill_features <- bill_main %>%
+     mutate(
+         combo_flag = ifelse(Combo_Opted == "Yes", 1, 0)
+     ) %>%
+     select(BillNo, Number_of_Products, Total_Amount, combo_flag)
>
> # Remove any missing values
> bill_features_clean <- bill_features %>%
+     filter(complete.cases(.))
>
> # Scale numeric features for k-means
> features_scaled <- bill_features_clean %>%
+     select(Number_of_Products, Total_Amount, combo_flag) %>%
+     scale()
>
> set.seed(123)  # for reproducibility
>
> # Choose k (example: 3 clusters)
> k <- 3
> kmeans_model <- kmeans(features_scaled, centers = k, nstart = 25)
>
> # Attach cluster label back to bills
> bill_segments <- bill_features_clean %>%
+     mutate(cluster = kmeans_model$cluster)
>
> head(bill_segments)
# A tibble: 6 × 5
  BillNo Number_of_Products Total_Amount combo_flag
```

```
  <dbl>          <dbl>      <dbl>     <dbl>
1     1            20       2780        0
2     2            21       2331        1
3     3            10       1220        0
4     4            10       2290        0
5     5            22       5236        1
6     6            18       4068        1
# ℹ 1 more variable: cluster <int>
>
> # Quick profile of each segment
> segment_profile <- bill_segments %>%
+   group_by(cluster) %>%
+   summarise(
+     n_bills        = n(),
+     avg_items      = mean(Number_of_Products),
+     avg_bill_value  = mean(Total_Amount),
+     combo_ratio    = mean(combo_flag),
+     .groups = "drop"
+   )
> segment_profile
# A tibble: 3 × 5
  cluster n_bills avg_items avg_bill_value
    <int>   <int>     <dbl>         <dbl>
1       1   10377      23.7         4326.
2       2    8459      12.1         2002.
3       3    8456      12.0         1977.
# ℹ 1 more variable: combo_ratio <dbl>
```

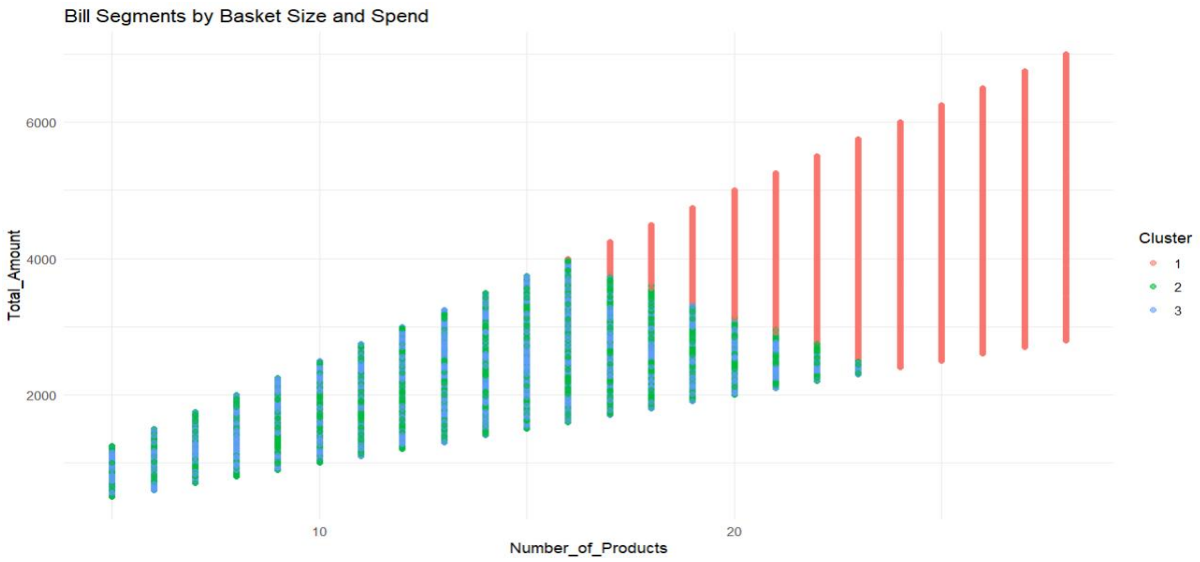

Figure 21. Bill Segments by Basket Size and Spend

2.6.18 MARKET BASKET RULES (APRIORI) FOR COMBO SUGGESTIONS

This section applies Market Basket Analysis using the Apriori algorithm to uncover meaningful associations between products frequently purchased together, helping identify strong bundle opportunities for future combo offers. Each bill is converted into a transaction containing unique product names, and a transaction matrix of 27,292 bills and 551 products is generated. The Apriori model is then run with a minimum support of 0.5 percent and a confidence threshold of 40 percent to extract rules that reveal which items consistently co-occur in customer baskets. Although no rules met the specified thresholds in this initial run, the process establishes a structured framework for exploring product affinities, enabling the business to refine parameters or adjust support levels to discover actionable product pairings that can guide smarter combo creation, cross-selling strategies, and category-level merchandising decisions.

Coding :

```
> library(ggplot2)
>
> ggplot(bill_segments,
+        aes(x = Number_of_Products, y = Total_Amount,
+            color = factor(cluster))) +
+     geom_point(alpha = 0.6) +
+     labs(title = "Bill Segments by Basket Size and Spend",
+          color = "Cluster") +
+     theme_minimal()
>
> library(dplyr)
> library(arules)
> library(dplyr)
> library(arules)
> # Prepare item list per bill: use Product_Name as item
> basket_df <- bill_product %>%
+     select(BillNo, Product_Name) %>%
+     distinct()   # avoid duplicate same product in same bill
>
> # Split into list of character vectors, one per BillNo
> basket_list <- split(basket_df$Product_Name, basket_df$BillNo)
>
> # Convert to 'transactions' object
> trans <- as(basket_list, "transactions")
> summary(trans)
transactions as itemMatrix in sparse format with
 27292 rows (elements/itemsets/transactions) and
 551 columns (items) and a density of 0.02942194

most frequent items:
```

```
      Maaretti Bruschette Roasted Garlic Bread 70 g
                                   1572
                          Mahali Potato 1Kg
                                   891
                 PPS Black Urad Dhal Split 500g
                                   880
         Colgate Extra Clean Medium Toothbrush 1 pc
                                   875
Whisper Bindazzz Nights Koala Soft XXL + 5 Pads 1 No
                                   875
                                (Other)
                                 437351

element (itemset/transaction) length distribution:
sizes
   3    4    5    6    7    8    9   10   11   12
   1   24 1162 1158 1108 1121 1196 1163 1205 1132
  13   14   15   16   17   18   19   20   21   22
1192 1224 1194 1120 1107 1212 1173 1165 1175 1178
  23   24   25   26   27   28
1201 1173 1169 1158 1021  560

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   3.00   10.00   16.00   16.21   22.00   28.00

includes extended item information - examples:
                                 labels
1 24 Mantra Organic Creamy Peanut Butter 450 g
2                      Aachi Appalam 100 gm
3                  Aachi Chilli Powder 50 g

includes extended transaction information - examples:
  transactionID
1             1
2             2
3             3
```

Run Apriori and inspect rules

```
> # Basic rules: items that are frequently bought together
> rules <- apriori(
+     trans,
+     parameter = list(
+         supp = 0.005,   # 0.5% minimum support (tune based on data size)
+         conf = 0.4,     # 40% minimum confidence
+         minlen = 2
+     )
+ )
Apriori
```

```
Parameter specification:
 confidence minval smax arem  aval originalSupport
        0.4    0.1    1 none FALSE            TRUE
 maxtime support minlen maxlen target  ext
       5   0.005      2     10  rules TRUE

Algorithmic control:
 filter tree heap memopt load sort verbose
    0.1 TRUE TRUE  FALSE TRUE    2    TRUE

Absolute minimum support count: 136

set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[551 item(s), 27292 transaction(s)] done [0.10s].
sorting and recoding items ... [551 item(s)] done [0.01s].
creating transaction tree ... done [0.03s].
checking subsets of size 1 2 done [0.02s].
writing ... [0 rule(s)] done [0.00s].
creating S4 object  ... done [0.01s].
```

CHAPTER III

3. LEARNING EXPERIENCES

3.1 REAL-TIME BUSINESS UNDERSTANDING

Real-time business understanding refers to the ability to interpret ongoing operations, customer behaviour, and revenue patterns as they happen, enabling managers to make informed decisions with minimal delay. In the context of supermarket analytics, real-time insights play a crucial role in responding quickly to demand fluctuations, consumer preferences, pricing reactions, and promotional effectiveness. For example, daily transaction data can show how combo offers influence customer purchasing behaviour, which price bands attract the highest sales volume, or whether particular categories (such as fruits, vegetables, or household essentials) are demonstrating sudden spikes or drop-offs. Understanding these patterns in real time helps management adjust stock levels, refine promotional strategies, prevent out-of-stock situations, and optimise discount-based sales.

Supermarkets operate with highly dynamic customer behaviour. Customers often make spontaneous decisions influenced by product visibility, pricing, seasonality, and peer choices. Real-time understanding allows businesses to immediately identify which products are moving faster than expected and which items are at risk of becoming slow movers. When integrated with inventory tracking, this data prevents overstocking, which ties up capital and increases wastage—especially for perishable items. For instance, a sudden rise in demand for fruits during festival months, or increased sales of sanitary products near month-end cycles, can inform precise procurement planning.

Real-time behavioural tracking also helps identify customer segments based on their spending patterns, basket sizes, and preference for combo or non-combo products. Such segmentation assists in targeted marketing, personalised recommendations, and loyalty programme planning. For instance, clusters of high-value shoppers who frequently purchase 20+ items may be ideal candidates for exclusive high-margin combo deals, while regular low-value customers may respond better to price-sensitive offers.

Additionally, real-time business understanding supports loss prevention. By tracking anomalies in product returns, cancellation rates, and billing inconsistencies, businesses can rapidly catch discrepancies or fraudulent activities. The ability to detect unusual purchasing behaviour—such as repeated bulk purchases of specific discounted items—can help management intervene before losses escalate.

Pricing decisions also benefit greatly from real-time analysis. As customers respond differently to various price bands, understanding demand elasticity helps supermarkets adjust prices strategically during peak versus off-peak hours or seasons. For example, if real-time data shows that price-band 0–50 consistently accounts for 42 percent of all line items, managers can focus promotions in this range to drive footfall while maintaining healthy margins on mid- and high-value items.

Overall, real-time business understanding transforms raw data into actionable intelligence. It reduces decision-making lag, enhances operational responsiveness, strengthens customer satisfaction, and supports efficient resource allocation. Supermarkets that integrate real-time dashboards with analytics tools gain a competitive advantage by staying ahead of consumer demand, preventing stock issues, reducing wastage, and refining revenue strategies. In essence, real-time understanding becomes the backbone of modern retail business intelligence, enabling continuous adaptation and long-term growth.

3.2 TECHNOLOGY EXPERIENCE (R, EDA)

Working with R programming and Exploratory Data Analysis (EDA) provides a powerful technical foundation for deriving insights from supermarket datasets. R is particularly effective for handling structured data, performing statistical operations, running forecasting models, generating visualisations, and implementing machine learning algorithms. The language offers an extensive collection of libraries—such as dplyr, ggplot2, forecast, prophet, and arules—that simplify complex analytical tasks and help analysts understand the business context with greater clarity.

The process begins with data cleaning, which includes handling missing values, removing duplicates, converting timestamps, and standardising categorical variables. For supermarket billing datasets, the cleaning phase is crucial because transaction-level inconsistencies—like incorrect product names, missing price entries, or mismatched category labels—can distort analysis. R’s data manipulation capabilities allow efficient transformation of messy data into a reliable analytical base using functions like mutate(), summarise(), filter(), and group_by().

Exploratory Data Analysis enhances understanding by uncovering hidden patterns within millions of line items. For instance, EDA enables analysts to calculate distribution metrics such as mean, median, quartiles, and standard deviations for product prices, total bill amounts, and item counts. Using graphical tools like histograms, bar plots, box plots, and line

charts, analysts can visually assess customer behaviour trends—such as which months exhibit higher combo conversion rates or which price bands dominate purchase frequency.

R also supports time-series analysis, which plays a vital role in understanding revenue stability and forecast modelling. With libraries like forecast and prophet, analysts can convert monthly revenue data into time-series objects and apply models like ARIMA or Prophet to anticipate future demand. These forecasts help businesses prepare inventory strategies, plan promotions, and manage cash flows efficiently.

Furthermore, R facilitates machine learning tasks like clustering. Through k-means clustering, bills can be segmented based on factors such as basket size, spending power, and combo adoption. This segmentation helps supermarkets understand diverse customer profiles—high-value, moderate, and budget-conscious shoppers.

Market Basket Analysis, using the arules package, enables discovery of frequent product pairings. Apriori rules reveal which products are commonly purchased together, guiding combo creation and store layout decisions. For example, if certain spices and bread products frequently appear together, supermarkets can place them closer, create targeted combos, or offer relevant promotions.

Overall, R and EDA provide a seamless technical workflow that connects data cleaning, transformation, visualisation, modelling, and machine learning. This holistic process ensures analysts can derive meaningful insights that are accurate, timely, and aligned with real business needs. The technology experience equips analysts with the capability to navigate large datasets, perform statistical reasoning, and turn raw numbers into strategic actions that drive supermarket performance.

3.3 INSIGHTS SUPPORT SUPERMARKET DECISIONS

Analytics-derived insights play a critical role in improving the operational efficiency, profitability, and customer experience of a supermarket. Data-driven decisions help managers understand product performance, shopper behaviour, pricing sensitivity, inventory risk, and promotional effectiveness. When structured effectively, insights from R-based EDA and modelling help supermarkets shift from reactive decisions to proactive, predictive strategies.

One major area where insights support decisions is inventory management. By analysing total quantity sold and identifying slow movers, supermarkets can manage stock levels more accurately. Product-level metrics like combo share, seasonal demand, or

dependency on discounts help determine whether to discontinue, promote, or reposition items on shelves. Forecasting models like ARIMA and Prophet provide future demand projections, preventing overstocking during low seasons and ensuring adequate supply during peaks.

Insights also inform pricing strategy. Price band distribution analysis reveals which ranges dominate sales. For example, if more than 80 percent of purchases fall within ₹0–200, management can design value-led pricing strategies to maximise conversions. Store-wide discount policies can be optimised by understanding the impact of price changes on customer behaviour.

Promotions and combo strategies benefit significantly from analytics. By identifying frequently purchased product pairs and measuring combo conversion rates, supermarkets can design bundles that align with natural buyer preferences rather than guesswork. Insights from Market Basket Analysis and k-means segmentation allow targeted combo design based on shopper types—high-value customers may respond to premium combos, while budget shoppers may prefer essential combos.

Customer insights further support personalisation efforts. By analysing bill-level data, supermarkets can classify shoppers into segments and tailor communication. High-value clusters can be offered loyalty rewards or premium service experiences, while moderate shoppers can receive email or SMS reminders about discounts.

Seasonal patterns uncovered through time-series analysis deepen the understanding of when combo effectiveness peaks. During festival seasons or monthly salary periods, supermarkets can increase promotions or stock up on items that experience surge demand.

Overall, data-driven insights transform supermarket decision-making across stocking, pricing, marketing, customer retention, and operational efficiency. They enable accurate forecasting, strategic planning, and improved profitability while reducing waste, increasing shopper satisfaction, and supporting sustainable business growth.

3.4 PRACTICAL PROBLEMS FACED DURING DATA COLLECTION

Data collection in a supermarket environment presents numerous practical challenges that directly affect the quality, accuracy, and completeness of analytical outcomes. One major issue concerns inconsistent data entry. Cashiers, suppliers, and inventory staff may use different naming conventions for the same product, leading to mismatched categories or

duplicate entries. For example, “Colgate Medium” and “Colgate Toothbrush Medium” may be separate entries even though they represent the same SKU.

Another challenge is missing data, especially in bill records where product quantities, prices, or combo flags may be unrecorded due to system glitches or human oversight. Missing timestamps or incorrect bill dates disrupt time-series analysis and delay accurate forecasting.

Physical product barcodes may also be damaged, leading to inaccurate scans or manual entries. Manual inputs are prone to errors like typing mistakes, leading to data inconsistencies. Differences between ERP systems, PoS systems, and accounting databases create challenges during data merging, causing duplication or mismatches.

Supermarkets also face high transaction volumes, with thousands of daily line items, making data storage, backup, and retrieval a significant operational burden. When transactional data is extracted from multiple systems, integration delays may occur.

Another major problem is categorisation inconsistency. Product categories may change over time due to rebranding, size changes, or pack variations. Keeping all items consistently mapped to correct categories is labour-intensive.

From a customer behaviour perspective, data collection may not fully capture non-digitised purchases such as cash payments that are recorded manually in some stores, creating gaps between actual and logged sales.

Seasonality, promotions, and sudden changes—like festival demand spikes—also distort baseline patterns, making the collected data volatile.

Despite these challenges, data cleansing, standardisation, and proper version control help mitigate errors. Consistent naming protocols, integrated POS systems, and automated categorisation pipelines can significantly improve data reliability for analysis.

3.5 ETHICAL / DATA PRIVACY LEARNING

Working with supermarket billing and customer-level data brings important ethical and data privacy responsibilities. The most fundamental principle is protecting customer identity. Even if the dataset contains only bill numbers and not personal identifiers, inferred patterns—such as repeated behaviours—can still reveal sensitive consumption habits. This means analysts must avoid attempts to re-identify individuals and ensure compliance with India's privacy standards and global frameworks like GDPR.

Ethical handling requires data minimisation, collecting only the details necessary for analysis and avoiding storage of personally sensitive information such as phone numbers, addresses, or payment details unless absolutely essential. When such information exists, encryption and access control mechanisms must be enforced.

Data security forms another pillar of ethical learning. Raw billing data must be stored securely, protected with encryption, and shared only through controlled channels. Cleaned datasets used for analysis should be stripped of unnecessary identifiers. Analysts must maintain confidentiality and ensure data is accessible only to authorised personnel.

Consent and transparency are also important. Supermarkets must clearly inform customers, typically through POS notifications or policy documents, that their anonymised purchase data may be used for improving operations or marketing.

Ethically, analysts must refrain from biased interpretations. For example, segmentation models should not stereotype customers or unfairly target vulnerable groups. Data-driven decisions must be justifiable, transparent, and focused on customer value rather than exploitation.

Another key learning is avoiding overfitting or misuse of predictive models. Forecasting and recommendation algorithms must be validated to prevent incorrect assumptions that may lead to customer dissatisfaction or misleading promotions.

Finally, ethical practice involves maintaining integrity in reporting—ensuring results are accurate, not manipulated, and clearly conveying limitations of the analysis. Through these principles, data privacy and ethical responsibility become central to trustworthy, sustainable, and customer-centric analytics.

CHAPTER - IV

4. CONCLUSION

4.1 KEY FINDINGS

- Combo purchases form a significant share of customer transactions, with the conversion rate consistently remaining between 48 percent and 53 percent across months. This indicates strong customer acceptance of combo offerings and demonstrates steady behavioural consistency across seasons.
- Revenue contribution from combos and non-combos is nearly equal, with combo bills accounting for 50.34 percent of total revenue and non-combos contributing 49.66 percent. Despite lower average bill value in combo bills, the higher volume of combo transactions balances overall contribution.
- Average bill value differs noticeably: non-combo bills show an average value of ₹2,836 while combo bills average ₹1,918. This suggests that combo-based purchases attract cost-sensitive buyers who respond positively to bundled savings.
- Price-band analysis reveals strong affinity towards lower-value products, with 42 percent of all line items falling under the ₹0–50 price range and almost 85 percent of purchases within the ₹0–200 bracket. This highlights a high concentration of demand in value-driven price segments.
- Price-band behaviour is identical for combo and non-combo bills, showing that combos do not artificially push customers towards cheaper items. Both groups maintain similar structure, indicating parallel buying tendencies.
- Product category distribution shows dominance of essential categories, with Fruits & Vegetables (21.7%), Household Essentials (18.1%), and Feminine Care (18.1%) leading category contribution. These categories anchor customer footfall and ensure stable daily revenue.
- Medium and niche categories show significant scope for uplift, such as Staples (5.7%), Kitchen Essentials (5.74%), and Baby Needs (2.33%), which can be strengthened through targeted combo design and shelf optimisation.
- Frequently paired product analysis reveals strong natural affinities, indicating high cross-sell potential. These relationships help supermarkets design meaningful bundles aligned with customer buying logic.

- Category-wise Combo Lift analysis shows that most categories generate 24–26 percent of their line items through combos, meaning customers adopt combos consistently across diverse segments without category bias.
- Product-level combo share analysis shows that certain SKUs rely heavily on combos for movement, with many products showing a combo-share between 24 and 30 percent of total quantity sold. These products benefit strongly from discount-driven exposure.
- Inventory risk analysis identifies slow-moving SKUs that still appear in combos, indicating that combos are helping clear low-velocity items, but such products require closer monitoring to avoid overstock risks.
- Combo Discount Dependency Index shows that no major SKU exceeds a 60 percent dependency level, meaning products are not critically over-dependent on combo discounts. However, several items show 28–30 percent dependency, signalling moderate reliance.
- Seasonal pattern analysis shows combo effectiveness remains stable, without drastic drops across months. Slight peaks during February, March, April, November, and September imply influence of festivals, salary cycles, or monthly shopping behaviour.
- Time-series forecasting using ARIMA shows stable, predictable combo revenue, averaging around ₹13.17 lakh per month with minimal fluctuation. This stability supports reliable procurement and stock planning.
- Prophet forecasting reveals a slow upward trend in combo revenue, along with notable seasonal waves. This means combination of trend and seasonality can be leveraged to plan promotions strategically during high-impact months.
- K-means clustering identifies three distinct customer segments, showing clear behavioural differences: high-value bulk purchasers, average-value routine shoppers, and low-value price-focused shoppers.
- Cluster 1 (high-value shoppers) displays the largest basket size and highest combos, making them ideal for premium bundles and loyalty-driven offers.
- Clusters 2 and 3 (routine shoppers) show lower combo adoption, suggesting scope for targeted awareness-driven campaigns or low-value essential combos.
- Market Basket Analysis provides structural groundwork but initial rules did not meet support thresholds. Adjusting parameters or expanding dataset size may reveal more meaningful cross-sell rules.

- Repeat Purchase Influence analysis shows that 551 products were first bought through combos, and later purchased without combos. This demonstrates strong trial-to-repeat conversion capacity of combo offerings.
- Combo purchases therefore act as an effective product introduction mechanism, particularly for new or low-visibility items, enhancing long-term brand penetration.
- The supermarket's customer base demonstrates hybrid purchasing behaviour, with buyers choosing combos for savings but returning to non-combos for routine restocking.
- Combo offerings remain a strategic advantage, particularly in a market dominated by low-price sensitivity and essential-category consumption.

4.2 BUSINESS RECOMMENDATIONS

A. Stock Planning

1. Maintain higher inventory for top categories like Fruits & Vegetables, Household Essentials, and Feminine Care due to consistent high footing.
2. Monitor slow-moving SKUs identified in the risk analysis and reduce purchase frequency while using combos strategically to clear excess inventory.
3. Leverage forecasting outputs to stock ahead for seasonal spikes—especially in Feb–Mar–Apr and Nov–Dec months.
4. Maintain balanced stock for mid-range categories to prevent demand gaps during combo promotions.
5. Introduce automated reorder triggers based on historical monthly combo revenue stability.

B. Discount Optimisation

6. Deploy targeted discounts only for items showing moderate discount dependency (28–30%) rather than blanket promotions.
7. Reduce discount intensity for SKUs showing strong natural repeat purchases.
8. Offer stackable or limited-time discounts during peak combo-conversion months identified through the Prophet model.
9. Introduce differentiated discounts for clusters: higher-value shoppers need value-add combos, while moderate shoppers need price-based combos.
10. Avoid over-discounting in categories already showing high organic demand.

C. Cross-Sell Strategy

11. Use frequently paired product insights to create natural cross-sell placements on shelves.
12. Position high-repeat combo items near complementary essentials to strengthen attach rates.
13. Re-run Apriori with lower thresholds to extract actionable cross-sell rules for combo creation.
14. Personalise cross-sell recommendations based on customer cluster profiles.
15. Introduce “Smart Combos” that combine high-velocity items with medium-velocity items to balance inventory turnover.

4.3 SCOPE FOR FURTHER RESEARCH

1. Extend Market Basket Analysis using more advanced algorithms (FP-Growth, Association Networks) to generate deeper combo suggestions.
2. Incorporate customer demographic data (age, location, loyalty membership) for richer segmentation and personalised recommendation modelling.
3. Expand forecasting models to include external factors such as festivals, climate cycles, and competitor discount trends.
4. Build a dynamic price elasticity model to understand how different customer clusters react to price changes.
5. Explore deep-learning-based demand forecasting (LSTM, GRU) for more accurate long-term combo revenue predictions.

CHAPTER - V

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